Highlights

Multi-Task Particle Swarm Optimization with Knowledge-Guided Elite Refinement and Two-Phase Weight Scheduling for High-Dimensional Feature Selection

Anonymous Author

- Closed-loop knowledge-guided swarm learning for high-D feature selection.
- MI seeding with adaptive recovery prevents stagnation and restores diversity.
- Knowledge-guided elite refinement accumulates feature signals with decay.
- Two-phase weight scheduling decouples feature pruning from accuracy tuning.
- KDS-EL achieves 95.74% accuracy with only 87.8 features on 20 benchmarks.

Multi-Task Particle Swarm Optimization with Knowledge-Guided Elite Refinement and Two-Phase Weight Scheduling for High-Dimensional Feature Selection

Anonymous Author[a,*]

[a] *Anonymous Institution (masked for double-blind review)*

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ABSTRACT

Feature selection in high-dimensional data remains a central challenge in evolutionary machine learning. Particle swarm optimization suffers from three structural deficiencies: undirected initialization provides insufficient search guidance in sparse high-dimensional spaces, uniform feature perturbation fails to exploit cross-iteration feature-importance feedback, and static objective weighting cannot adapt to shifting evolutionary demands. To overcome these bottlenecks, this paper proposes Knowledge-Guided Dual-Phase Swarm Evolutionary Learning (KDS-EL), a closed-loop framework incorporating domain prior injection, generational knowledge accumulation, and dynamic objective scheduling. Specifically, mutual-information-guided seeding provides continuous relevance priors while adaptive re-initialization restores population diversity upon stagnation. The knowledge-guided elite refinement strategy dynamically harvests cross-iteration feature feedback with exponential memory decay, steering elite solutions toward historically effective dimensions under bounded computational overhead. Furthermore, two-phase weight scheduling temporally decouples early coarse dimensionality reduction from late-stage accuracy refinement. Extensive evaluations across 20 ultra-high-dimensional benchmark datasets demonstrate that KDS-EL achieves a leading average classification accuracy of 95.74% while isolating an ultra-sparse signature of only 87.8 features, significantly outperforming 11 representative competitor algorithms and foundational multi-task frameworks. Non-parametric statistical tests with Holm–Bonferroni correction confirm the statistically significant superiority and computational efficiency of the proposed framework.

1. Introduction

High-dimensional data have emerged as a fundamental challenge in classification problems across diverse domains, including genomics, text mining, and computer vision. These datasets typically contain hundreds or even tens of thousands of features, while only a limited number of labeled samples are available, resulting in the well-known $p \gg n$ scenario. Under such conditions, conventional learning algorithms often suffer from severe overfitting, increased computational complexity, and reduced interpretability due to the curse of dimensionality [1, 2]. Feature selection (FS) has therefore become an essential preprocessing technique that aims to identify a compact and informative subset of features while preserving classification performance. However, selecting an optimal feature subset remains highly challenging because the search space grows exponentially with the number of features D, and the underlying subset-selection problem is NP-hard [3]. Moreover, individual feature relevance measures are insufficient to characterize complex feature interactions and dependencies, which are often critical for achieving robust classification performance.

FS methods are generally categorized into filter, wrapper, and embedded approaches [1, 4]. Filter methods rank features according to statistical criteria and offer high computational efficiency, but they usually evaluate features independently of the classifier and may overlook feature interactions relevant to classification. Embedded methods incorporate feature selection into the learning process, providing a compromise between selection efficiency and predictive performance; however, their effectiveness is often dependent on the characteristics of specific learning models. Wrapper methods directly optimize feature subsets according to classifier performance and can achieve competitive classification results, but the repeated model evaluations make them computationally expensive, especially in high-dimensional search spaces. To address the combinatorial complexity of wrapper-based FS, evolutionary computation has been extensively investigated as a promising optimization paradigm [3, 5–7]. By maintaining a population of candidate solutions, evolutionary algorithms can perform global exploration of feature subsets without requiring gradient information. Among them, particle swarm optimization (PSO), genetic algorithms, and differential evolution have demonstrated promising results on high-dimensional FS tasks.

PSO has become one of the most widely adopted evolutionary algorithms for FS due to its simple structure, efficient information-sharing mechanism, and fast convergence. Nevertheless, conventional PSO-based FS methods still encounter three structural limitations when dealing with sparse and high-dimensional search spaces. First, random initialization distributes particles without prior guidance throughout the binary search space, whereas effective feature subsets usually occupy only a small region with limited active dimensions. Consequently, a considerable number of fitness evaluations may be wasted on redundant and suboptimal solutions before the swarm identifies promising search regions. Second, conventional uniform feature perturbation strategies treat all dimensions equally and neglect the feature-level improvement information accumulated during previous evaluations, preventing the search from exploiting historically informative features. Third, existing PSO-based FS methods typically employ fixed objective weights to balance classification error and feature subset size during the entire optimization process. However, the desired trade-off between feature reduction and accuracy refinement varies across search stages: aggressive sparsity enhancement is preferred in early exploration, whereas accurate decision boundary refinement becomes more important after a compact subset has been obtained.

Recent evolutionary FS studies have attempted to improve search performance through enhanced population initialization, adaptive evolutionary operators, and refined fitness evaluation strategies [8–11]. Although these approaches have achieved promising results, they usually focus on isolated components of the optimization process and do not jointly address inefficient initialization, insufficient exploitation of accumulated feature knowledge, and inflexible objective weighting. Recently, multi-task PSO has emerged as an effective paradigm for high-dimensional FS with fixed-length binary representations. For instance, MEL [12] adopts cooperative subpopulations and a fixed-length representation to efficiently handle feature spaces containing tens of thousands of dimensions. Despite its effectiveness, existing multi-task PSO-based FS frameworks still leave several challenges unresolved: the initial population is generally generated without sufficient feature-level prior knowledge, useful feature-importance information discovered during evolution is not explicitly retained and exploited, and the balance between classification performance and feature sparsity remains unchanged across different search stages.

To address these limitations, this paper proposes Knowledge-Guided Dual-Phase Swarm Evolutionary Learning (KDS-EL), a closed-loop framework that cohesively integrates domain prior injection, cross-iteration knowledge accumulation, and dynamic objective scheduling into a cooperative multi-task PSO backbone.

The main contributions of this paper are summarized as follows:

1. *MI-Guided Initialization with Adaptive Re-initialization*: A hybrid seeding strategy that injects continuous feature-relevance priors into population construction to eliminate early-stage search waste, coupled with an adaptive re-initialization mechanism that restores swarm diversity upon evolutionary stagnation (Section 3.2).
2. *Knowledge-Guided Elite Refinement (KER)*: A targeted elite refinement strategy that dynamically harvests cross-iteration feature-importance feedback with exponential memory decay, directing elite solutions toward historically effective features within a strictly bounded evaluation overhead (Section 3.3).

*Corresponding author.
E-mail address: corresponding.author@example.com (Anonymous Author).
ORCID(s):

Table 1: Methodological taxonomy of investigated evolutionary feature selection algorithms.

		Knowledge Engineering			Search Control		
Algorithm	Paradigm	MI Prior Seeding	Online Memory	Memory Decay	Elite Refinement	Dynamic Weighting	Complexity
BPSO [7]	SO-SI	×	×	×	×	×	High
FWPSO [8]	SO-SI	○	×	×	×	×	Moderate
SaWDE [9]	SO-DE	×	×	×	×	○	High
IGPSO [10]	SO-SI	○	×	×	×	×	Moderate
PSO-CSM [11]	SO-SI	○	×	×	×	×	Moderate
DENCA-SO [14]	SO-DE	○	×	×	×	×	Low
bIRIME [13]	SO-SI	×	×	×	○	×	High
BinCOA [15]	SO-SI	×	×	×	○	×	High
MEL [12]	MT-SI	×	✓	×	×	×	High
KDS-EL (Proposed)	**MT-SI**	✓	✓	✓	✓	✓	**Low**

✓: Fully supported; ○: Partially supported; ×: Absent / not supported.
Paradigms: SO: Single-Objective; MT: Multi-Task; SI: Swarm Intelligence; DE: Differential Evolution.

3. *Two-Phase Weight Scheduling (TPS)*: A stage-aware objective scheduling mechanism that temporally decouples early aggressive feature pruning from late-stage accuracy refinement, overcoming the rigidity of static weighting (Section 3.4).
4. *Comprehensive Empirical Validation*: Extensive evaluations across 20 high-dimensional benchmark datasets (spanning 500 to 54,675 features) against state-of-the-art single-objective competitor algorithms and the original MEL framework, rigorously validated through non-parametric statistical testing, component contribution analysis, and parametric sensitivity investigations (Section 4).

Extensive experiments demonstrate that KDS-EL achieves the highest classification performance, obtaining an average classification accuracy of 0.9574 with only 87.8 selected features, corresponding to an average feature selection ratio of merely 1.1%. Compared with recent top-tier competitor bIRIME [13] and established competitor FWPSO [8], KDS-EL improves classification accuracy by 1.20 and 3.97 percentage points, while reducing the selected feature count by 78.5% and 91.8%, respectively. Furthermore, KDS-EL achieves a 3.4× speedup over the original MEL algorithm [12], reducing average runtime from 175.0 s to 51.3 s. Pairwise Wilcoxon signed-rank tests with Holm–Bonferroni step-down correction confirm that the performance advantages are statistically significant over all peer competitor algorithms and intermediate configurations.

The remainder of this paper is organized as follows. Section 2 formulates the feature selection optimization problem and reviews related literature and background concepts. Section 3 presents the proposed KDS-EL framework and formulates its three constituent mechanisms in detail. Section 4 presents comprehensive experimental results, statistical tests, component contribution analysis, and parameter sensitivity investigations. Finally, Section 5 concludes the paper and discusses prospective research directions.

2. Related Work and Background

This section establishes the formal mathematical foundation of feature selection and systematically reviews the relevant evolutionary computation paradigms. Section 2.1 first formalizes the combinatorial optimization problem and categorizes classical search paradigms. Section 2.2 subsequently analyzes evolutionary swarm mechanics alongside the structural failure modes of classical stochastic binarization in high-dimensional spaces. To address high-dimensional search challenges, Section 2.3 scrutinizes the cooperative multi-task evolutionary learning framework and pinpoints its operational bottlenecks. Finally, Section 2.4 investigates knowledge-guided evolutionary search, contrasts offline filter priors against online closed-loop memory, and establishes the taxonomic positioning of KDS-EL in contrast to existing literature.

2.1. Problem formulation and search paradigms in feature selection

The essence of feature selection lies in combinatorial optimization, aiming to identify an optimal, minimal subset of informative features from a high-dimensional feature space to maximize classification performance while eliminating redundant and uninformative dimensions [1, 2]. Formally, let $\mathcal{D} = \{(\mathbf{f}_i, y_i)\}_{i=1}^{N_{\text{samp}}}$ denote a supervised training dataset comprising N_{samp} instances, where $\mathbf{f}_i \in \mathbb{R}^D$ represents the full feature vector in the original D-dimensional space and $y_i \in \{1, 2, \ldots, C\}$ designates the discrete class label across C categories. In discrete binary representations, a candidate feature subset is formulated as a binary decision indicator vector $\mathbf{x} \in \{0, 1\}^D$, defined as:

$$x_j = \begin{cases} 1, & \text{if feature } f_j \text{ is selected,} \\ 0, & \text{if feature } f_j \text{ is discarded.} \end{cases} \tag{1}$$

The primary objective of feature selection is to simultaneously minimize the classification error rate and the cardinality of the selected feature subset. In evolutionary wrapper feature selection, this multi-criterion goal is mathematically formalized as minimizing a scalarized single-objective fitness function $f(\mathbf{x})$ [16, 17]:

$$\min_{\mathbf{x}\in\{0,1\}^D} f(\mathbf{x}) = w_a \cdot \text{ErrorRate}(\mathbf{x}) + w_f \cdot \left(\frac{\|\mathbf{x}\|_0}{D}\right), \tag{2}$$

subject to $\mathbf{x} \in \{0, 1\}^D$ and $w_a + w_f = 1$, where ErrorRate($\mathbf{x}$) represents the cross-validation classification error evaluated on the masked feature subset $\{\mathbf{f}_i \odot \mathbf{x}\}$ using an inductive k-nearest neighbors (k-NN) learner, with $\odot$ denoting the Hadamard element-wise product. The term $\|\mathbf{x}\|_0$ denotes the number of active features, $\|\mathbf{x}\|_0/D$ represents the feature selection ratio, and $w_a, w_f \in [0, 1]$ serve as trade-off coefficients balancing predictive accuracy against subset compactness. Because the cardinality of the discrete search space is 2^D, evaluating all candidate subsets exhaustively is NP-hard and computationally intractable when $D \gg 10^3$. Traditional feature selection techniques are broadly categorized into three classical paradigms based on their evaluation criteria: filter, embedded, and wrapper methods [2, 4]. Specifically, *filter methods* evaluate feature subsets based solely on intrinsic statistical or information-theoretic metrics—such as mutual information in mRMR [18], correlation in FCBF [19], or centrality scores [20]—independently of any classification algorithm. While filter methods are computationally lightweight with $O(DN_{\text{samp}} + D \log D)$ complexity, their classifier-agnostic evaluation and rigid hard truncation permanently discard weakly relevant features that exhibit strong collective synergy when combined. Conversely, *embedded methods* perform feature selection natively during classifier training by embedding sparsity constraints into model parameter estimation, such as L_1-regularized regression in Bolasso [21] or tree-based ensemble importance in FS-GBDT [22]. Although embedded methods bridge computational efficiency and predictive feedback, they remain architecturally bound to specific learners and frequently encounter performance degradation under severe sample scarcity where $N_{\text{samp}} \ll 100$. In contrast, *wrapper methods* utilize a target learning algorithm as a black-box evaluator, assessing candidate feature masks directly through cross-validation predictive error feedback [4, 16]. By tightly coupling subset evaluation with classifier feedback, wrappers naturally capture complex non-linear feature interactions and yield superior classification accuracy.

To navigate the discrete 2^D combinatorial space in wrapper feature selection without gradient information, Evolutionary Computation paradigms—including Genetic Algorithms [23], Ant Colony Optimization [24], Artificial Bee Colony [25], and Swarm Intelligence [3, 26]—have emerged as dominant search heuristics. To combat the curse of dimensionality in small-sample high-dimensional data, various advanced single-objective evolutionary strategies have been introduced. For instance, self-adaptive differential evolution with adaptive weighting (SaWDE) [9] dynamically

balances classification error against subset cardinality; feature-weight particle swarm optimization (FWPSO) [8] injects offline feature weights to steer velocity updates; importance-guided particle swarm optimization (IGPSO) [10] employs multi-layer perceptron importance scores to bias particle movements; and particle swarm optimization with comprehensive scoring mechanism (PSO-CSM) [11] leverages multi-criterion feature evaluation to restrict search trajectories. Despite these advancements, standard single-objective evolutionary wrappers face severe structural bottlenecks in ultra-high-dimensional spaces where $D \gg 10^3$. Uniform random initialization scatters individuals indiscriminately across the search space, generating overly dense candidate masks with approximately $D/2$ active features that burden classifier evaluations and provoke premature overfitting. Furthermore, stochastic variation operators treat feature perturbations as memoryless trials, discarding historical feature-importance feedback across generations. Finally, static or coarsely tuned objective weighting enforces rigid trade-offs throughout evolution, failing to adapt to the distinct requirements of early-stage dimensional reduction and late-stage accuracy refinement.

2.2. Evolutionary swarm intelligence and discrete feature selection dynamics

Standard particle swarm optimization, originally introduced by Kennedy and Eberhart [5], is a population-based stochastic meta-heuristic inspired by the collective foraging behaviors of bird flocks and fish schools. A swarm consists of N candidate particles navigating a D-dimensional continuous solution space $\mathbb{R}^D$. At iteration t, each particle i is characterized by a continuous position vector $\mathbf{p}_i^{(t)} \in \mathbb{R}^D$ and an associated velocity vector $\mathbf{v}_i^{(t)} \in \mathbb{R}^D$, steered dynamically by its personal historical best position $\mathbf{p}_{\text{best},i} \in \mathbb{R}^D$ and the swarm's global best position $\mathbf{g}_{\text{best}} \in \mathbb{R}^D$. The velocity and position updates of particle i at iteration $t+1$ are formulated as follows [6]:

$$\mathbf{v}_i^{(t+1)} = \omega \mathbf{v}_i^{(t)} + c_1 r_1 (\mathbf{p}_{\text{best},i} - \mathbf{p}_i^{(t)}) + c_2 r_2 (\mathbf{g}_{\text{best}} - \mathbf{p}_i^{(t)}), \tag{3}$$

$$\mathbf{p}_i^{(t+1)} = \mathbf{p}_i^{(t)} + \mathbf{v}_i^{(t+1)}, \tag{4}$$

where $\omega \in [0, 1]$ denotes the inertia weight controlling momentum retention, $c_1, c_2 > 0$ are the cognitive and social acceleration coefficients governing the attraction toward individual and collective best experiences, and $r_1, r_2 \sim \mathcal{U}(0, 1)$ are independent uniform random factors providing stochastic exploration capability.

To adapt continuous PSO to the combinatorial nature of binary feature selection where candidate solutions are represented as binary indicator masks $\mathbf{x}_i \in \{0, 1\}^D$, Kennedy and Eberhart [7] proposed Binary PSO (BPSO). In BPSO, continuous velocities are mapped into selection probabilities via a Sigmoid transfer function:

$$S(v_{i,j}^{(t+1)}) = \frac{1}{1 + \exp(-v_{i,j}^{(t+1)})}, \tag{5}$$

and the corresponding binary decision variable $x_{i,j}^{(t+1)}$ is sampled probabilistically as:

$$x_{i,j}^{(t+1)} = \begin{cases} 1, & \text{if } r_{i,j} < S(v_{i,j}^{(t+1)}), \\ 0, & \text{otherwise}, \end{cases} \tag{6}$$

where $r_{i,j} \sim \mathcal{U}(0, 1)$ is an independent uniform random threshold. Although BPSO exhibits fast convergence on low-dimensional benchmarks, it encounters severe exploration-exploitation dilemmas when applied to ultra-high-dimensional biomedical domains where $D \gg 10^3$. Because the Sigmoid function satisfies $S(0) = 0.5$, continuous velocities hovering near zero inevitably yield an expected active feature density of approximately $D/2$. In typical biomedical datasets containing $D \geq 10{,}000$ dimensions, this stochastic mapping generates candidate subsets with more than 5,000 active features, heavily overloading wrapper classifier evaluations with $O(N_{\text{samp}}^2 d_{\text{selected}})$ complexity and triggering severe overfitting under small sample sizes where $N_{\text{samp}} \ll 100$. Furthermore, continuous stochastic sampling introduces high generational variance: even when velocity vectors stabilize, candidate feature masks continue to jitter stochastically, causing particle trajectories to degrade into memoryless random walks across the combinatorial space.

To overcome the random drift of standard binarization in high-dimensional feature selection, diverse single-objective meta-heuristic algorithms have recently been proposed. Evolutionary filter methods, such as DENCA-SO [14], employ neighborhood component feature selection (NCFS) continuous weights within differential evolution operators. By evaluating candidate subsets using algebraic surrogate costs rather than wrapper classifiers, DENCA-SO achieves rapid wall-clock execution; however, because its optimization process remains entirely decoupled from inductive classification margins, it suffers from severe feature redundancy and retains over 48% of original dimensions on biomedical benchmarks. Concurrently, advanced binary swarm optimizers seek to enhance discrete exploration. For instance, bIRIME [13] incorporates dynamic escape energy, composite mutation strategies, and opposition-based restart mechanisms to mitigate premature convergence, whereas BinCOA [15] couples refractive opposition-based learning with crossed horizontal and vertical search operations. Nevertheless, these multi-operator strategies incur prohibitive computational burdens: generating and evaluating multiple trial vectors per agent scales wrapper classifier calls to three or four times the population size per generation, causing execution times to escalate beyond 2,000 seconds on large-sample benchmarks (Section 4.2.6). Crucially, in ultra-high dimensions without proactive pruning guidance, stochastic binarization traps these algorithms near the uninformative 50% dimensionality attractor, preserving thousands of redundant features. On the other hand, wrapper methods employing scalarized fitness formulations—such as SaWDE [9], FWPSO [8], and PSO-CSM [11]—enforce fixed static objective weights throughout evolution. This static weighting creates a fundamental evolutionary phase misalignment: the search process inherently requires aggressive feature pruning during early exploration to eliminate vast noise subspaces, whereas delicate decision-boundary fine-tuning is required during late-stage exploitation to maximize predictive accuracy.

2.3. Multi-task evolutionary learning and the MEL framework

Evolutionary Multi-Task Optimization (EMTO) aims to solve multiple distinct optimization tasks simultaneously by facilitating inter-task knowledge transfer through a unified chromosomal representation [27]. In feature selection, EMTO frameworks partition the global population into collaborative subpopulations to explore complementary search subspaces, including minion-assisted multi-tasking [28], evolutionary multi-task PSO for hyperspectral and gene expression data [29, 30], multi-task bee colony band selection [31], and large-scale evolutionary multi-task optimization [32]. Most notably, Wang et al. [12] introduced Multi-Task Evolutionary Learning (MEL), which establishes a dual-subpopulation PSO framework. In MEL, a population of N particles is partitioned into two collaborative subpopulations $\mathcal{P}_1$ and $\mathcal{P}_2$, with each subpopulation comprising $N/2$ individuals operating on complementary search tasks. Specifically, *Subpopulation 1* executes continuous swarm exploration, updating continuous velocities and positions using PSO dynamics augmented with an inter-task knowledge transfer channel directing particles toward the historical global best of Subpopulation 2 ($\mathbf{g}_{\text{best},2} \in \mathbb{R}^D$):

$$\mathbf{v}_{1,i}^{(t+1)} = \omega \mathbf{v}_{1,i}^{(t)} + c_1 r_1 (\mathbf{p}_{\text{best},1,i} - \mathbf{p}_{1,i}^{(t)}) + c_2 r_2 (\mathbf{g}_{\text{best},1} - \mathbf{p}_{1,i}^{(t)}) + c_3 r_3 (\mathbf{g}_{\text{best},2} - \mathbf{p}_{1,i}^{(t)}), \tag{7}$$

where $\omega \in [0, 1]$ is the inertia weight, $c_1, c_2, c_3 > 0$ are acceleration coefficients governing self-cognition, social learning, and cross-task transfer, and $r_1, r_2, r_3 \sim \mathcal{U}(0, 1)$ are independent stochastic factors. Concurrently, *Subpopulation 2* performs importance-guided discrete sampling directly from a shared feature-importance matrix $\mathbf{W}_{\text{MEL}} \in \mathbb{R}^D$, whose elements accumulate the historical selection frequencies of features across successful candidate solutions.

Both subpopulations evaluate candidate subsets using a shared scalar fitness function similar to Eq. (2) that combines classification error and feature subset ratio via fixed static trade-off weights $\alpha, \beta \in [0, 1]$ satisfying $\alpha + \beta = 1$.

While MEL demonstrates the utility of cross-task knowledge transfer, its search dynamics in ultra-high-dimensional spaces where $D \gg 10^3$ suffer from critical structural bottlenecks. Continuous particles in $\mathcal{P}_1$ are initialized uniformly across the unit hypercube, lacking domain-informed guidance toward sparse biomarker subspaces and wasting substantial evaluation budget exploring irrelevant dimensions in early iterations. Moreover, the importance matrix $\mathbf{W}_{\text{MEL}}$ accumulates historical feature frequencies unconditionally without exponential forgetting or elite fitness verification, allowing obsolete early choices to permanently bias subsequent discrete sampling in $\mathcal{P}_2$ and trapping the swarm in local optima. In addition, rigid static weighting prevents the algorithm from dynamically adjusting its selection pressure between early aggressive feature pruning and late high-precision boundary refinement. The proposed KDS-EL framework directly adopts this dual-subpopulation multi-task foundation while resolving each of these structural deficiencies through targeted mechanisms.

2.4. Knowledge-guided search and taxonomic positioning

Integrating domain-specific prior knowledge into meta-heuristics has attracted surging attention as an effective means to escape the blindness of black-box search [17, 33, 34]. In high-dimensional feature selection, Mutual Information (MI), rooted in Shannon's information theory [35], serves as the primary information-theoretic metric to quantify non-linear feature relevance. For continuous feature variables $f_j \in \mathbb{R}$ and discrete class labels $Y \in \{1, \ldots, C\}$, the

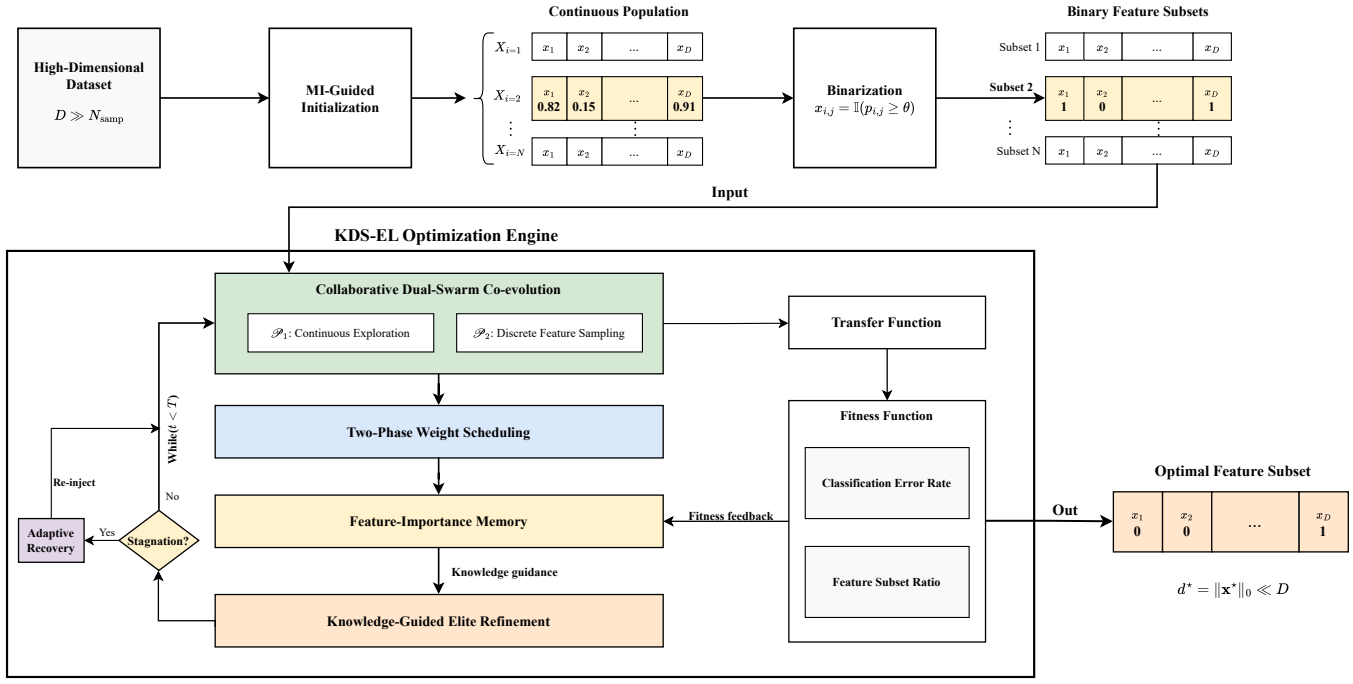


Figure 1: Overall dataflow architecture of the proposed KDS-EL framework for high-dimensional feature selection. The pipeline progresses from mutual-information-guided prior initialization to collaborative dual-swarm co-evolution ($\mathcal{P}_1$ and $\mathcal{P}_2$), dynamic Two-Phase Weight Scheduling, and closed-loop elite refinement with exponential memory decay, isolating optimal compact biomarker subsets.

mutual information $I(f_j; Y)$ is formulated as:

$$I(f_j; Y) = H(f_j) - H(f_j \mid Y) = \sum_{c=1}^{C} P(Y = c) \int_{\mathbb{R}} p(f_j \mid Y = c) \log_2 \frac{p(f_j \mid Y = c)}{p(f_j)} \, df_j, \tag{8}$$

where $p(f_j)$ and $p(f_j \mid Y = c)$ denote the marginal and class-conditional probability density functions, respectively. In empirical applications where true underlying distributions are unknown, calculating continuous mutual information through naive data binning introduces severe bin-width sensitivity and estimation bias. Instead, non-parametric entropy estimation based on k-nearest neighbors distance statistics—specifically the Kraskov–Stögbauer–Grassberger (KSG) estimator [36] extended to mixed continuous-discrete systems by Ross [37]—directly evaluates nonlinear dependency without requiring parametric assumptions. Conventional filter feature selection methods [17, 18] evaluate mutual information across all D dimensions and apply a rigid hard truncation that retains only the top-k ranked features, permanently discarding the remainder. Despite its computational speed of $O(DN_{\text{samp}} + D \log D)$, hard truncation fundamentally ignores multi-feature synergy: individual features with moderate scores that are collectively decisive when combined are permanently eliminated from the search space.

Existing knowledge-guided evolutionary algorithms typically adopt one-shot offline injection via either static filter pre-filtering or static velocity biasing. In static filter pre-filtering, exemplified by MI-MPODE [34] and NNS-DFPG-CSA [17], algorithms compute feature rankings offline and initialize the population or restrict the search space to the top-k ranked features. In static velocity biasing, such as FWPSO [8], static feature-weight vectors are incorporated to bias particle velocity equations throughout evolution. While beneficial, one-shot knowledge injection lacks online adaptability: once injected, the static prior is never updated or refined by the empirical trial-and-error feedback generated during subsequent wrapper evaluations. Conversely, learning-based surrogates, such as neural networks [10] or Gaussian processes [38], introduce severe surrogate training overhead and risk model misspecification under sample-scarce conditions where $N_{\text{samp}} \ll 100$.

To overcome the collective limitations identified in the existing literature, KDS-EL establishes a closed-loop, knowledge-guided evolutionary learning framework that unifies offline domain priors, online memory harvesting, and stage-aware objective scheduling. Specifically, normalized mutual information scores are embedded as continuous Gaussian-perturbed seeds, preserving the full search space while guiding the swarm toward an ultra-sparse initial basin, complemented by an adaptive re-initialization mechanism to recover exploratory diversity upon stagnation (Section 3.2). Throughout evolution, cross-iteration fitness feedback is harvested into a cumulative importance matrix $\mathbf{W} \in \mathbb{R}^D$ governed by exponential forgetting, enabling the KER strategy to execute targeted, directional elite flips within a strictly bounded computational budget (Section 3.3). Finally, a dynamic Two-Phase Weight Scheduling strategy temporally decouples early coarse feature pruning from late high-precision accuracy refinement, achieving simultaneous gains in classification accuracy, feature compactness, and runtime efficiency (Section 3.4). Table 1 systematically summarizes the methodological taxonomy and structural comparison between KDS-EL and all nine competitor feature selection algorithms investigated in this study across seven orthogonal technical dimensions.

3. The Proposed Method

To address the severe challenges of ultra-high-dimensional feature selection ($D \gg 10^3, N_{\text{samp}} \ll 100$), this section presents the proposed KDS-EL framework in detail. Specifically, the macro data-flow architecture and algorithmic flowchart of KDS-EL are outlined in Section 3.1. The prior-guided sparse initialization and adaptive recovery strategies are detailed in Section 3.2. Section 3.3 analyzes the search failure of conventional random perturbations, and formulates the closed-loop mechanism of cumulative importance learning and knowledge-guided elite refinement. The dynamic two-phase weight scheduling strategy is presented in Section 3.4. Finally, the overall algorithmic procedure and its computational complexity analysis are synthesized in Section 3.5.

3.1. The framework of KDS-EL

The macro data-flow architecture and problem mapping of KDS-EL for high-dimensional feature selection are systematically illustrated in Fig. 1. Given a high-dimensional dataset $\mathcal{D}$, the search pipeline begins by extracting filter-level mutual information to construct a continuous prior relevance landscape. An initial swarm of continuous positions is generated via hybrid prior seeding and deterministically mapped into sparse binary candidate masks. Throughout optimization, these candidate subsets are transferred to the KDS-EL optimization engine, which coordinates collaborative dual-swarm co-evolution, dynamic two-phase fitness evaluations, cumulative feature-importance harvesting, and targeted elite refinement in closed-loop synergy with a wrapper classification evaluation engine. Upon meeting the convergence criterion, the framework outputs the optimal compact biomarker subset $\mathbf{x}^\star$ alongside its validated classification performance.

The execution logic and operational flowchart across evolutionary generations are depicted in Fig. 2. In KDS-EL, a swarm of N candidate solutions

across D features is formally represented as an $N \times D$ binary decision matrix $\mathbf{X}$:

$$\mathbf{X} = \begin{bmatrix} \mathbf{x}_1 \\ \mathbf{x}_2 \\ \vdots \\ \mathbf{x}_N \end{bmatrix} = \begin{bmatrix} x_{1,1} & x_{1,2} & \cdots & x_{1,D} \\ x_{2,1} & x_{2,2} & \cdots & x_{2,D} \\ \vdots & \vdots & \ddots & \vdots \\ x_{N,1} & x_{N,2} & \cdots & x_{N,D} \end{bmatrix} \in \{0,1\}^{N \times D}, \tag{9}$$

where the i-th row vector $\mathbf{x}_i = [x_{i,1}, x_{i,2}, \ldots, x_{i,D}] \in \{0,1\}^D$ denotes an individual candidate solution, with $x_{i,j} = 1$ indicating that feature f_j is selected and $x_{i,j} = 0$ indicating exclusion.

To balance global exploration and local exploitation in sparse high-dimensional spaces, the population matrix $\mathbf{X}$ is partitioned into two collaborative subpopulations $\mathcal{P}_1$ and $\mathcal{P}_2$ of equal size. Subpopulation $\mathcal{P}_1$ explores continuous coordinates via velocity–position updates with an inter-task knowledge transfer channel guided by the global best solution of $\mathcal{P}_2$. Concurrently, $\mathcal{P}_2$ conducts discrete probability-driven feature sampling informed by the accumulated feature-importance landscape. Compared with conventional single-population architectures, this dual-subpopulation design establishes complementary search dynamics: continuous swarm movement preserves momentum to explore unvisited spaces, while discrete sampling focuses evaluations on verified high-relevance dimensions, effectively mitigating premature stagnation.

During iterative optimization, KDS-EL coordinates three complementary mechanisms into a closed loop. First, population initialization injects Gaussian-perturbed mutual information priors to generate sparse candidate masks, while an adaptive re-initialization mechanism recovers population diversity whenever search progress halts. Second, differential fitness improvements across generations are accumulated into a continuous importance matrix under exponential memory decay, which actively guides the KER strategy to perform targeted directional bit flips under a greedy one-to-one in-place replacement rule. Third, the dynamic TPS strategy temporally shifts objective weights across evolutionary stages, enforcing aggressive feature pruning in the early phase and fine-grained accuracy tuning in the late phase. By continuously cycling through these coordinated stages, KDS-EL systematically prunes non-informative dimensions while progressively refining compact biomarker signatures.

3.2. Population initialization and adaptive recovery

In ultra-high-dimensional feature selection, standard uniform random initialization activates each feature with equal probability, generating excessively dense candidate subsets with approximately half the features active ($d \approx D/2$). Evaluating these dense candidate subsets exhausts computational budgets on non-informative dimensions and traps search trajectories in noisy local basins. To establish an informative yet diverse initial population while mitigating mid-evolution search stagnation, KDS-EL introduces a filter-wrapper hybrid seeding strategy coupled with an adaptive stagnation recovery mechanism, as illustrated in Fig. 3.

3.2.1. Prior-guided sparse population initialization

Given the training dataset instance $\mathcal{D} = \{(\mathbf{f}_i, y_i)\}_{i=1}^{N_{\text{samp}}}$ with continuous feature vectors $\mathbf{f}_i = [f_{i,1}, \ldots, f_{i,D}] \in \mathbb{R}^D$ and discrete class labels $y_i \in \{1, \ldots, C\}$, the non-linear dependencies between individual candidate dimensions and target categories are quantified through continuous-discrete mutual information [37]. To handle continuous real-valued inputs without relying on arbitrary data discretization, the mutual information score $\hat{I}(f_j; Y)$ for each feature $j \in \{1, \ldots, D\}$ is computed via the non-parametric k-nearest neighbors KSG estimator [36, 37]:

$$\hat{I}(f_j; Y) = \psi(N_{\text{samp}}) - \frac{1}{N_{\text{samp}}} \sum_{i=1}^{N_{\text{samp}}} \psi(N_{y_i}) + \psi(k) - \frac{1}{N_{\text{samp}}} \sum_{i=1}^{N_{\text{samp}}} \psi(m_i), \tag{10}$$

where $\psi(\cdot)$ denotes the digamma function, N_{y_i} is the count of training samples sharing class label y_i, k represents the neighborhood size in the joint continuous-discrete space ($k = 3$), and m_i is the count of data points whose marginal continuous distance to sample i along feature f_j is strictly smaller than the distance to its k-th neighbor in the class-conditioned subspace. To establish a standardized probability profile across heterogeneous datasets, these raw mutual information scores are normalized into a prior relevance vector $\mathbf{m} = [m_1, \ldots, m_D]^T \in [0,1]^D$:

$$m_j = \frac{\hat{I}(f_j; Y) - \min_{1 \le k \le D} \hat{I}(f_k; Y)}{\max_{1 \le k \le D} \hat{I}(f_k; Y) - \min_{1 \le k \le D} \hat{I}(f_k; Y) + \epsilon}, \quad j = 1, \ldots, D, \tag{11}$$

where $\epsilon > 0$ is a small numerical constant preventing division by zero.

To guarantee strict methodological integrity and eliminate selection bias, the filter prior calculation is architecturally restricted to the training partition presented to the optimizer. In an inductive machine learning deployment where an external unseen test partition $\mathcal{D}_{\text{test}}$ or an outer nested evaluation loop is prescribed, the prior relevance vector $\mathbf{m}$ and the subsequent evolutionary wrapper search are executed strictly on the training partition $\mathcal{D}_{\text{train}}$, ensuring that unseen validation samples remain completely unobserved throughout all search stages. In the standardized benchmark optimization protocol adopted in this work (Section 4.1), $\mathcal{D}$ serves as the complete benchmark problem instance, within which an internal 5-fold cross-validation scheme functions as the wrapper fitness evaluation surrogate.

To prevent population homogenization while guiding search trajectories toward informative dimensions, candidate particles are initialized through a hybrid seeding scheme parameterized by a seeding ratio $\rho \in (0,1)$. Continuous position vectors $\mathbf{p}_i = [p_{i,1}, \ldots, p_{i,D}] \in [0,1]^D$ are constructed as:

$$\mathbf{p}_i = \begin{cases} \text{clip}\left(\mathbf{m} + \boldsymbol{\varepsilon}_i,\ 0,\ 1\right), & 1 \le i \le \lfloor \rho N \rfloor, \\ \mathbf{u}_i, & \lfloor \rho N \rfloor < i \le N, \end{cases} \tag{12}$$

where $\boldsymbol{\varepsilon}_i \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I}_D)$ represents zero-mean Gaussian perturbation with standard deviation σ, $\mathbf{u}_i \sim \mathcal{U}(0,1)^D$ provides uniform stochastic coverage across unguided dimensions, and $\text{clip}(z, 0, 1) = \max(0, \min(1, z))$ restricts continuous coordinates to the unit hypercube.

Continuous position vectors $\mathbf{p}_i$ are subsequently transformed into binary decision vectors $\mathbf{x}_i = [x_{i,1}, \ldots, x_{i,D}] \in \{0,1\}^D$ via a deterministic Heaviside step mapping with threshold $\theta \in (0.5, 1)$:

$$x_{i,j} = \begin{cases} 1, & \text{if } p_{i,j} \ge \theta, \\ 0, & \text{otherwise}, \end{cases} \quad j = 1, \ldots, D. \tag{13}$$

By setting the cut-off threshold $\theta > 0.50$, only features possessing above-average prior relevance are initialized in active states. This produces an intrinsically sparse initial population that biases exploration toward high-potential biomarker candidates, thereby resolving the cold-start bottleneck, reducing early evaluation latency, and accelerating convergence toward promising feature subspaces.

Estimation robustness and mitigation of the Winner's curse under sample scarcity: In ultra-high-dimensional and sample-scarce regimes where $D \gg 10^3$ and $N_{\text{samp}} \ll 100$ (such as *Adenoma* with $N_{\text{samp}} = 36, D = 7{,}399$ or *ALL4* with $N_{\text{samp}} = 63, D = 12{,}600$), non-parametric entropy estimation on limited observations inherently exhibits non-trivial finite-sample variance. Under extensive dimensional testing, stochastic sample alignment across thousands of candidate features may cause irrelevant noise dimensions to obtain spuriously elevated mutual information estimates—a classic manifestation of the Winner's curse in high-dimensional screening. If subjected to standard Min-Max normalization, such estimation fluctuations risk being amplified toward $m_j \approx 1$, potentially injecting false-positive biases into the initial population.

Crucially, KDS-EL establishes a multi-layered architectural defense that insulates the evolutionary search against noisy prior contamination. First, KDS-EL fundamentally differs from conventional filter pre-selectors by rejecting irreversible hard feature truncation; the normalized vector $\mathbf{m}$ acts solely as a continuous soft prior, leaving the entire $\mathbb{R}^D$ feature space fully accessible throughout search. Second, the hybrid initialization in (12) injects Gaussian dispersion $\boldsymbol{\varepsilon}_i \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I}_D)$ around $\mathbf{m}$ while maintaining an unguided sub-swarm $\mathbf{u}_i \sim \mathcal{U}(0,1)^D$ across $(1-\rho)N$ particles, preventing premature population collapse into false-positive empirical peaks. Third, every candidate subset is immediately subjected to wrapper cross-validation classification feedback; because spurious noise features fail to generalize across cross-validation folds, they generate zero or negative fitness differentials. Under the exponential memory decay governed by decay factor γ in (17), their transient weights in the cumulative importance matrix $\mathbf{W}$ rapidly decay to zero. Consequently, the knowledge-guided elite refinement mechanism swiftly purges these non-contributing dimensions in favor of synergistic, predictive biomarkers. Finally, should the swarm become temporarily trapped in empirical local attractors, the adaptive stagnation recovery routine (Section 3.2.2) re-samples the stagnant subpopulation under fresh stochastic perturbations, ensuring robust long-term exploratory self-correction.

3.2.2. Adaptive stagnation recovery

In multi-modal high-dimensional search spaces, population diversity inevitably decays across generations, which may lead the swarm into suboptimal attractors. To resolve this issue, KDS-EL incorporates an adaptive re-initialization routine triggered whenever the global-best fitness fails to improve for τ consecutive generations.

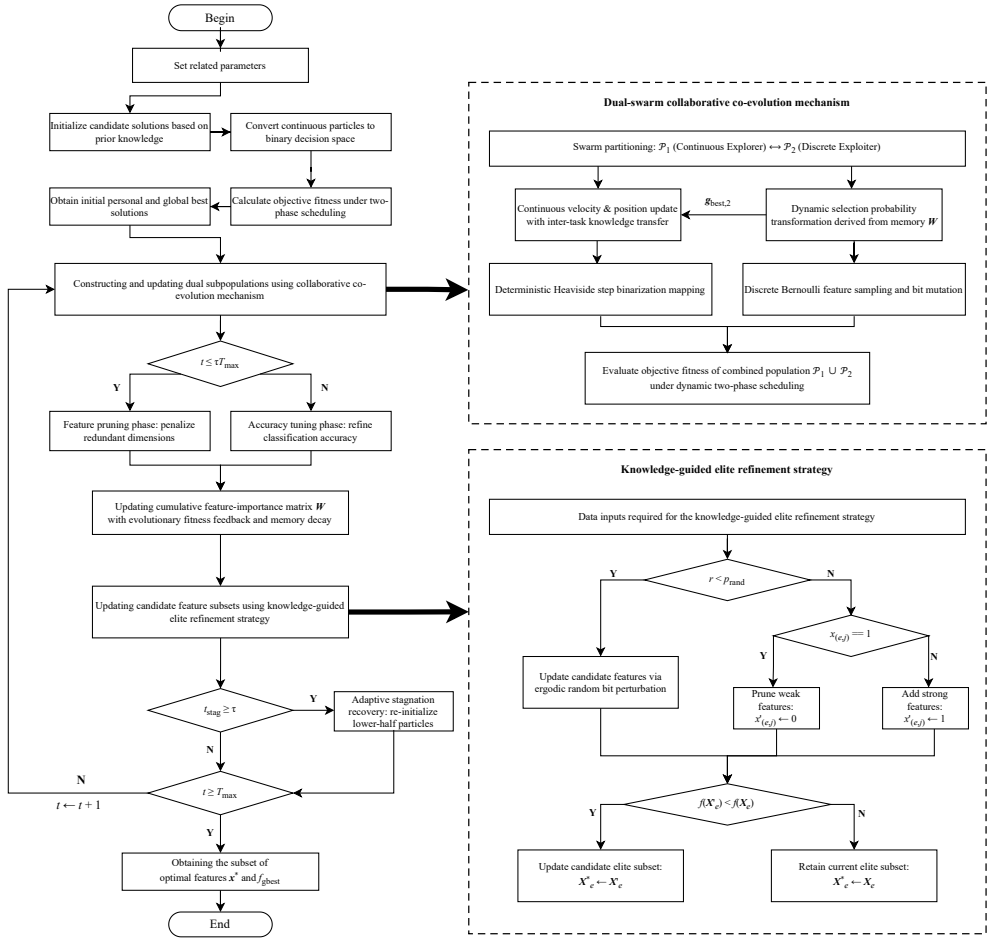


Figure 2: Algorithmic execution flowchart of the KDS-EL framework. The schematic illustrates the full optimization lifecycle, depicting parameter initialization, dual-subpopulation co-evolutionary updates, two-phase objective fitness evaluation, adaptive diversity recovery upon search stagnation, and closed-loop knowledge-guided elite refinement.

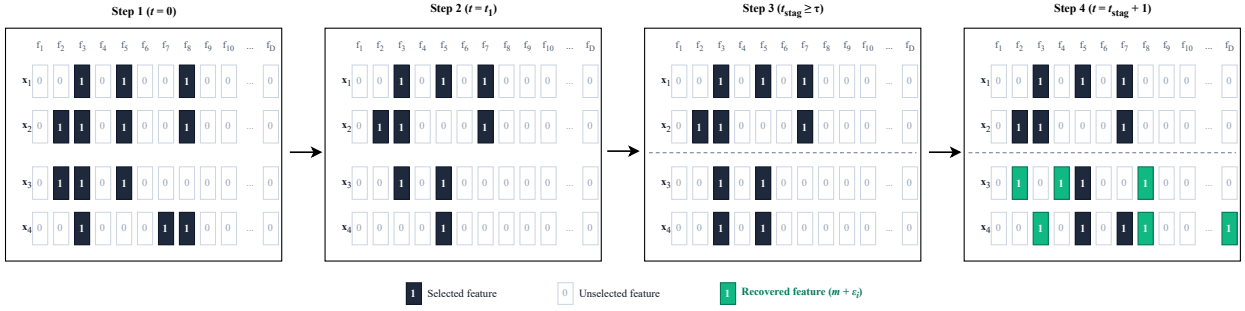


Figure 3: Schematic representation of prior-guided sparse population initialization and adaptive diversity recovery. Step 1 injects mutual information priors into continuous candidate vectors, Steps 2–3 trace generational convergence toward compact subsets, and Step 4 triggers adaptive re-initialization upon reaching the stagnation threshold τ to restore swarm diversity.

As depicted in Fig. 3, particles are first sorted in ascending order of their personal-best fitness values. The superior half of the swarm is retained intact to safeguard historical search trajectories and verified biomarker combinations. Concurrently, the lower-performing half is refreshed by re-sampling continuous positions around the prior relevance vector $\mathbf{m}$ under Gaussian perturbation:

$$\mathbf{p}_i \leftarrow \text{clip}\,(\mathbf{m} + \boldsymbol{\varepsilon}_i,\ 0,\ 1)\,, \quad x_{i,j} \leftarrow \mathbb{I}(p_{i,j} \ge \theta), \quad \forall\, j = 1, \ldots, D, \tag{14}$$

where $\mathbb{I}(\cdot)$ denotes the indicator function. The refreshed particles are immediately re-evaluated under current objective weights, and the stagnation counter is reset. By combining elite preservation with targeted prior re-injection, this mechanism restores exploratory diversity within high-potential subspaces without discarding accumulated optimization experience.

3.3. Cumulative importance learning and elite refinement

To overcome the search amnesia and destructive perturbation bottlenecks of conventional meta-heuristics, KDS-EL establishes a closed-loop knowledge learning mechanism comprising online importance accumulation and targeted directional elite refinement.

In ultra-high-dimensional feature selection ($D \gg 10^3$), traditional evolutionary wrappers rely on undirected stochastic bit flips with a uniform mutation probability $p_{\text{mut}} = 1/D$. Under this blind perturbation regime, meta-heuristics inevitably suffer from two fatal structural search lesions. First, when an established, highly informative biomarker ($x_{e,j} = 1$) is blindly mutated to 0, the delicate multi-feature synergy is abruptly dismantled, inducing severe classification degradation. Second, when an uninformative background noise feature ($x_{e,j} = 0$) is randomly flipped to 1, the candidate subset expands unnecessarily, escalating distance evaluation latency and precipitating classifier overfitting. Because the probability of hitting the few decisive dimensions within a 2^D combinatorial neighborhood is infinitesimal ($O(1/D) \approx 10^{-4}$), undirected mutation squanders the vast majority of function evaluations on destructive trials. Furthermore, conventional meta-heuristics exhibit severe search amnesia, as historical fitness feedback from intermediate iterations is discarded immedi-

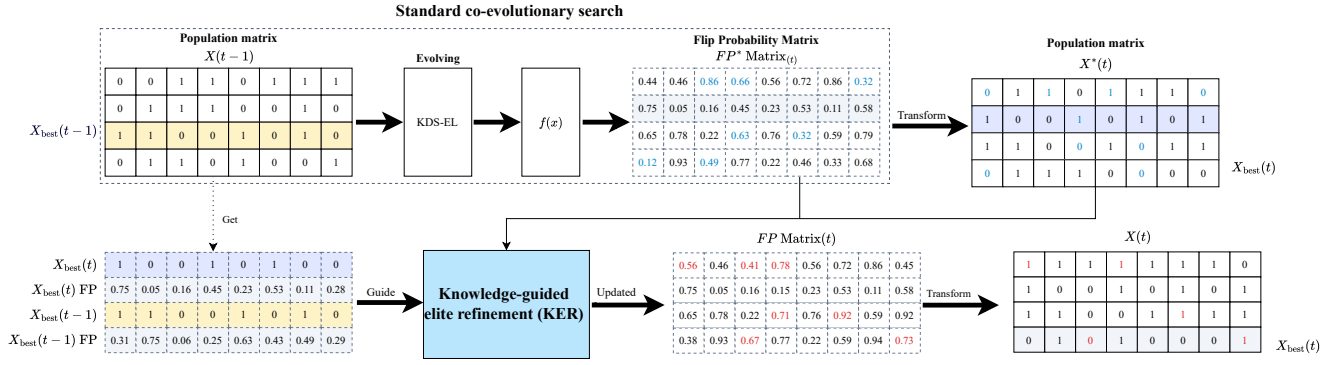


Figure 4: Operational mechanism of the knowledge-guided elite refinement strategy. Cross-iteration fitness differentials are harvested into a continuous feature-importance vector under exponential memory decay, generating a normalized bit-flip probability matrix that steers targeted directional mutations on the top-k_e elite solutions via greedy in-place replacement.

ately after particle position updates.

3.3.1. Online feature importance learning

To systematically harvest search experience across generations, KDS-EL maintains a continuous cumulative feature-importance vector $\mathbf{W} = [W_1, \ldots, W_D]^T \in \mathbb{R}^D$, initialized to $\mathbf{0}$.

At generation t, whenever a candidate particle $\mathbf{x}_i^{(t)}$ achieves a strict fitness improvement over its historical personal best $\mathbf{p}_{\text{best},i}$, the differential feature set $C_i^{(t)}$ comprising features whose selection status changed is extracted:

$$C_i^{(t)} = \left\{ j \in \{1, \ldots, D\} \;\middle|\; x_{i,j}^{(t)} \neq p_{\text{best},i,j} \right\}. \tag{15}$$

Let $\delta_i^{(t)} = f(\mathbf{p}_{\text{best},i}, \mathbf{w}(t)) - f(\mathbf{x}_i^{(t)}, \mathbf{w}(t)) > 0$ denote the scalar fitness improvement. Each altered feature $j \in C_i^{(t)}$ receives a directional reward proportional to the magnitude of $\delta_i^{(t)}$:

$$W_j \leftarrow W_j + \delta_i^{(t)} \cdot (2x_{i,j}^{(t)} - 1), \quad \forall j \in C_i^{(t)}, \tag{16}$$

where the term $(2x_{i,j}^{(t)} - 1)$ yields $+1$ if feature j was newly activated and -1 if deactivated. Candidate solutions that fail to improve fitness ($\delta_i^{(t)} \leq 0$) leave $\mathbf{W}$ unchanged.

To prevent early-discovered features from dominating the importance landscape indefinitely and maintain sensitivity toward emergent synergistic biomarker interactions, an exponential memory decay is applied at the conclusion of each generation:

$$\mathbf{W} \leftarrow \gamma \cdot \mathbf{W}, \tag{17}$$

where $\gamma \in (0, 1)$ is the memory retention factor. Consequently, features that consistently contribute to fitness improvements sustain high positive scores, whereas transiently selected or uninformative dimensions exponentially fade toward zero.

3.3.2. Knowledge-guided elite refinement

Elite individuals represent the most promising feature subsets discovered by the swarm; however, they may still retain redundant dimensions or lack subtle synergistic biomarkers. Rather than blind perturbation, the KER strategy exploits the continuous importance landscape $\mathbf{W}$ to guide local mutation toward high-potential dimensions under bounded computation ($k_e \ll N$).

At each generation, the top k_e individuals ranked by fitness are selected into the elite pool. For each elite candidate $\mathbf{x}_e \in \{0, 1\}^D$, the number of perturbed features is set to $k_{\text{flip}} = \max(1, \lfloor \phi D \rfloor)$, where $\phi \in (0, 1)$ represents the perturbation fraction. As depicted in the data-flow diagram in Fig. 4, KER probabilistically executes one of two operational pathways:

Branch 1: Ergodic random exploration (p_{rand}): To preserve search ergodicity and prevent the elite pool from stagnating in deceptive local optima, a feature index subset $\mathcal{S}_{\text{rand}} \subset \{1, \ldots, D\}$ of size k_{flip} is sampled uniformly at random, and the corresponding bits are inverted:

$$x'_{e,j} \leftarrow 1 - x_{e,j}, \quad \forall j \in \mathcal{S}_{\text{rand}}, \quad |\mathcal{S}_{\text{rand}}| = k_{\text{flip}}. \tag{18}$$

Branch 2: Knowledge-guided directional refinement ($1 - p_{\text{rand}}$): For each feature $j \in \{1, \ldots, D\}$, a state-dependent directional flip score σ_j is computed:

$$\sigma_j = (1 - 2x_{e,j}) \cdot W_j = \begin{cases} +W_j, & \text{if } x_{e,j} = 0, \\ -W_j, & \text{if } x_{e,j} = 1. \end{cases} \tag{19}$$

Algorithm 1 Knowledge-guided elite refinement

Require: Elite candidate solution $\mathbf{x}_e \in \{0, 1\}^D$, cumulative importance vector $\mathbf{W} \in \mathbb{R}^D$, perturbation fraction $\phi \in (0, 1)$, exploration probability $p_{\text{rand}} \in [0, 1]$, dynamic weight vector $\mathbf{w}(t) \in [0, 1]^2$
Ensure: Refined elite candidate solution $\mathbf{x}_e^\star \in \{0, 1\}^D$

```
k_flip ← max(1, ⌊φ · D⌋)
x'_e ← x_e                                   ▷ Initialize mutant candidate
Sample random variable r ~ U(0, 1)
if r < p_rand then                           ▷ Branch 1: Ergodic random exploration
    Sample random feature index subset S_rand ⊂ {1, ..., D} with |S_rand| = k_flip
    x'_{e,j} ← 1 − x_{e,j},   ∀j ∈ S_rand
else                                         ▷ Branch 2: Knowledge-guided directional refinement
    for j = 1 to D do
        σ_j ← (1 − 2x_{e,j}) · W_j           ▷ Compute directional flip score via (19)
    end for
    Identify index subset S_know of size k_flip with top σ_j scores via (20)
    x'_{e,j} ← 1 − x_{e,j},   ∀j ∈ S_know
end if
Evaluate candidate fitness f(x'_e, w(t)) via (23)
if f(x'_e, w(t)) < f(x_e, w(t)) then         ▷ Greedy pairwise replacement
    x*_e ← x'_e
else
    x*_e ← x_e
end if
return x*_e
```

A positive flip score $\sigma_j > 0$ indicates either an unselected feature ($x_{e,j} = 0$) with positive historical contribution ($W_j > 0$), or an active feature ($x_{e,j} = 1$) whose deactivation has historically improved fitness ($W_j < 0$). The feature subset $\mathcal{S}_{\text{know}}$ containing the k_{flip} features with the largest directional flip scores is identified:

$$\mathcal{S}_{\text{know}} = \arg \max_{\substack{\mathcal{S} \subseteq \{1, \ldots, D\} \\ |\mathcal{S}| = k_{\text{flip}}}} \sum_{j \in \mathcal{S}} \sigma_j, \tag{20}$$

and their binary states are inverted: $x'_{e,j} \leftarrow 1 - x_{e,j}, \forall j \in \mathcal{S}_{\text{know}}$.

Greedy one-to-one in-place replacement: The perturbed candidate $\mathbf{x}'_e$ is evaluated using the objective function $f(\mathbf{x}'_e, \mathbf{w}(t))$. A greedy acceptance rule determines solution admission:

$$\mathbf{x}_e^\star = \begin{cases} \mathbf{x}'_e, & \text{if } f(\mathbf{x}'_e, \mathbf{w}(t)) < f(\mathbf{x}_e, \mathbf{w}(t)), \\ \mathbf{x}_e, & \text{otherwise.} \end{cases} \tag{21}$$

Upon achieving a strict fitness improvement, $\mathbf{x}'_e$ immediately replaces $\mathbf{x}_e$ in the population and updates the global-best solution if $f(\mathbf{x}'_e, \mathbf{w}(t)) < f_{\text{gbest}}$. Because the elite pool size k_e is fixed ($k_e \ll N$), KER operates within an exact bounded overhead of k_e evaluations per generation. The complete procedural logic of KER is formalized in Algorithm 1.

Dual-scale search dynamics and dimension-scaling analysis: The perturbation budget $k_{\text{flip}} = \max(1, \lfloor \phi D \rfloor)$ scales linearly with ambient dimensionality D, producing distinct search behaviors across problem scales. In moderate-dimensional domains where $D \leq 2{,}000$, k_{flip} produces small-scale bit flips that execute fine-grained local neighborhood refinement around elite solutions. Conversely, in ultra-high-dimensional spaces where $D > 10^4$—such as the 54,675-dimensional *T1D* benchmark where $k_{\text{flip}} \approx 546$—KER shifts dynamically from localized perturbation into a guided macro-recombination mechanism. Operating at this broader structural scale allows the swarm to evaluate

coordinated multi-feature activations and deactivations guided by $\mathbf{W}$, facilitating transitions across distant combinatorial basins. Crucially, the greedy one-to-one in-place replacement rule in (21) serves as an intrinsic safeguard against destructive mutations: whenever an extensive macro-flip degrades classification accuracy or inflates feature penalties, the candidate mutant is discarded without modifying the elite pool. While fixed-fraction scaling $\lfloor \phi D \rfloor$ performs consistently across diverse benchmark scales, adaptively modulating the perturbation budget according to the active subset cardinality $d_{\text{selected}} = \|\mathbf{x}_e\|_0$ via $k_{\text{flip}} = \max(1, \lfloor \phi \cdot \|\mathbf{x}_e\|_0 \rfloor)$, or enforcing an upper saturation cap $k_{\max}$, represents an effective adaptive refinement for extreme-dimensional applications.

3.4. Dynamic two-phase weight scheduling

In high-dimensional feature selection, wrapper optimization must balance two fundamentally conflicting objectives: minimizing classification error and minimizing the selected feature subset cardinality. In the early evolutionary phase, the search space is dominated by an overwhelming number of redundant and noisy dimensions; thus, the primary optimization imperative is aggressive dimensionality pruning. In contrast, during the late evolutionary phase when search trajectories have converged to a compact subspace, the priority shifts to fine-tuning synergistic biomarker interactions to maximize classification accuracy. Conventional static weighting schemes with fixed trade-off coefficients fail to reconcile these temporally distinct demands, frequently inducing either inadequate early feature elimination or premature penalization of critical predictive features in later stages.

To resolve this temporal conflict, KDS-EL introduces the dynamic TPS strategy, which decouples the evolutionary horizon into two distinct functional phases. The total generation budget T is divided at a split point parameterized by the phase transition ratio $\kappa \in (0, 1)$. The dynamic objective weight vector $\mathbf{w}(t) = [w_a(t), w_f(t)] \in [0, 1]^2$, satisfying $w_a(t) + w_f(t) = 1$, is formulated as:

$$\mathbf{w}(t) = \begin{cases} [w_{a,1}, w_{f,1}], & 1 \le t \le \lfloor \kappa T \rfloor, \\ [w_{a,2}, w_{f,2}], & \lfloor \kappa T \rfloor < t \le T, \end{cases} \tag{22}$$

where the weight coefficients satisfy the ordering $w_{f,1} > w_{f,2}$ and $w_{a,1} < w_{a,2}$.

At generation t, the unified scalar fitness $f(\mathbf{x}, \mathbf{w}(t))$ of a binary feature subset $\mathbf{x} \in \{0, 1\}^D$ is evaluated as:

$$f(\mathbf{x}, \mathbf{w}(t)) = w_a(t) \cdot (1 - \text{Acc}(\mathbf{x})) + w_f(t) \cdot \left(\frac{\|\mathbf{x}\|_0}{D} \right), \tag{23}$$

where $\text{Acc}(\mathbf{x}) \in [0, 1]$ represents the 5-fold cross-validation classification accuracy evaluated by a k-NN classifier, and $\|\mathbf{x}\|_0 = \sum_{j=1}^{D} x_j$ denotes the ℓ_0-norm quantifying the number of active features.

During Phase 1 ($1 \le t \le \lfloor \kappa T \rfloor$), the elevated feature penalty $w_{f,1} = 0.30$ enforces severe selective pressure against non-informative features, rapidly compressing the active candidate dimensionality to a compact fraction of D. Subsequently, upon entering Phase 2 ($\lfloor \kappa T \rfloor < t \le T$), the weight vector transitions to prioritize classification accuracy ($w_{a,2} = 0.95$), empowering the swarm to explore subtle multi-gene synergies over the condensed feature subspace without being prematurely penalized for retaining essential predictive biomarkers.

3.5. Overall algorithmic procedure and complexity analysis

Algorithm 2 synthesizes the overall execution pipeline of KDS-EL into a structured multi-stage optimization framework. The complete workflow coordinates offline filter relevance extraction, knowledge-informed hybrid seeding, dynamic two-phase objective scheduling, collaborative dual-subpopulation search, differential importance harvesting, and targeted elite refinement.

3.5.1. Computational complexity analysis

The computational overhead of KDS-EL decouples into two distinct phases: one-time offline preprocessing and online generational evolution. In the offline phase, evaluating non-parametric mutual information scores across D candidate dimensions with N_{samp} training samples via k-nearest neighbors search requires $O(D \cdot N_{\text{samp}} \log N_{\text{samp}})$ operations, while normalizing and sorting these relevance estimates incurs $O(D \log D)$ time. Because this procedure is executed exactly once prior to the evolutionary search, the offline preprocessing complexity $\mathcal{T}_{\text{pre}} = O(D \cdot N_{\text{samp}} \log N_{\text{samp}} + D \log D)$ represents a negligible fraction of the total computational budget, especially in small-sample regimes where $N_{\text{samp}} \ll 100$. Throughout online evolution, each generation $t \in \{1, \ldots, T\}$ orchestrates three coordinated operational stages. First, subpopulation updates and cross-task knowledge transfer—comprising continuous velocity-position updates in $\mathcal{P}_1$ and discrete probabilistic sampling in $\mathcal{P}_2$—require $O(N \cdot D)$ arithmetic operations per generation. Second, wrapper fitness evaluation constitutes the dominant computational expenditure: for

Algorithm 2 The overall framework of KDS-EL

Require: Training dataset $\mathcal{D} = \{(\mathbf{f}_i, y_i)\}_{i=1}^{N_{\text{samp}}}$, population size $N \in \mathbb{Z}_+$, maximum generations $T \in \mathbb{Z}_+$, binarization threshold $\theta \in (0, 1)$, prior seeding ratio $\rho \in (0, 1)$, perturbation scale $\sigma \in \mathbb{R}_+$, forgetting factor $\gamma \in (0, 1)$, elite pool size $k_e \in \mathbb{Z}_+$, perturbation fraction $\phi \in (0, 1)$, random branch probability $p_{\text{rand}} \in [0, 1]$, phase transition split ratio $\kappa \in (0, 1)$, stagnation threshold $\tau \in \mathbb{Z}_+$

Ensure: Optimal compact feature subset $\mathbf{x}^\star \in \{0, 1\}^D$ and optimal fitness $f_{\text{gbest}} \in \mathbb{R}$

Stage 1: Filter-level relevance extraction

Compute normalized mutual information vector $\mathbf{m} \in [0, 1]^D$ via (10) and (11)

Stage 2: Knowledge-informed hybrid population seeding

Initialize population $\mathcal{P} = \{\mathbf{x}_i\}_{i=1}^{N}$ with $\lfloor \rho N \rfloor$ Gaussian MI seeds and $(N - \lfloor \rho N \rfloor)$ uniform particles via (12)–(13)

Initialize importance vector $\mathbf{W} \leftarrow \mathbf{0} \in \mathbb{R}^D$, stagnation counter $c_{\text{stall}} \leftarrow 0$, and initial weights $\mathbf{w}(1) = [w_{a,1}, w_{f,1}]$ via (22)

Evaluate initial candidate fitness values $\{f(\mathbf{x}_i, \mathbf{w}(1))\}_{i=1}^{N}$ via (23); initialize personal bests $\{\mathbf{p}_{\text{best},i}\}_{i=1}^{N}$ and global best $\mathbf{g}_{\text{best}}$

Stage 3: Dynamic two-phase co-evolutionary loop

for $t = 1$ **to** T **do**

  Update dynamic objective weight vector $\mathbf{w}(t)$ according to (22)

  Update continuous Subpopulation $\mathcal{P}_1$ via continuous PSO with inter-task knowledge transfer [12]

  Update discrete Subpopulation $\mathcal{P}_2$ via discrete probability sampling guided by $\mathbf{W}$

  Evaluate candidate solution fitness $\{f(\mathbf{x}_i, \mathbf{w}(t))\}_{i=1}^{N}$ using 5-fold cross-validation k-NN classifier via (23)

  Update cumulative importance vector $\mathbf{W}$ with improving differentials and apply memory decay γ via (16)–(17)

  Update personal bests $\{\mathbf{p}_{\text{best},i}\}_{i=1}^{N}$ and identify top k_e elite candidate indices $\mathcal{E}$

  for each $e \in \mathcal{E}$ **do** ▷ Knowledge-guided elite refinement

    $\mathbf{p}_{\text{best},e} \leftarrow \text{KER}(\mathbf{p}_{\text{best},e}, \mathbf{W}, \phi, p_{\text{rand}}, \mathbf{w}(t))$ ▷ Invoke Algorithm 1

  end for

  if $\min_i f(\mathbf{p}_{\text{best},i}, \mathbf{w}(t)) < f(\mathbf{g}_{\text{best}}, \mathbf{w}(t))$ **then**

    Update global best $\mathbf{g}_{\text{best}} \leftarrow \arg\min_i f(\mathbf{p}_{\text{best},i}, \mathbf{w}(t))$ and reset $c_{\text{stall}} \leftarrow 0$

  else

    Increment stagnation counter $c_{\text{stall}} \leftarrow c_{\text{stall}} + 1$

  end if

  if $c_{\text{stall}} \ge \tau$ **then** ▷ Adaptive stagnation recovery

    Refresh lower-performing half of population using Gaussian-perturbed MI seeds via (14)

    Re-evaluate refreshed particles and reset $c_{\text{stall}} \leftarrow 0$

  end if

end for

return $\mathbf{x}^\star \leftarrow \mathbf{g}_{\text{best}}$, $f_{\text{gbest}} \leftarrow f(\mathbf{g}_{\text{best}}, \mathbf{w}(T))$

each candidate solution $\mathbf{x}_i$ with active feature cardinality $d_i = \|\mathbf{x}_i\|_0$, computing pairwise Euclidean distances under 5-fold cross-validation with a k-NN classifier incurs $O(N_{\text{samp}}^2 \cdot d_i)$ runtime, aggregating to $O(N \cdot N_{\text{samp}}^2 \cdot \bar{d}^{(t)})$ across the swarm of size N, where $\bar{d}^{(t)} = \frac{1}{N} \sum_{i=1}^{N} d_i$ denotes the average active feature count at generation t. Third, cumulative importance learning and elite refinement introduce targeted search intensification within strictly bounded overhead: extracting differential feature masks and updating the continuous memory vector $\mathbf{W}$ consumes $O(N \cdot D)$ operations; computing directional flip scores across all D dimensions and isolating the top k_{flip} features incurs $O(D \log D)$ time via partial quick-sort; and re-evaluating the refined elite candidates requires $O(k_e \cdot N_{\text{samp}}^2 \cdot d_{\text{elite}})$ operations. Because the elite pool size is strictly bounded with $k_e \ll N$ ($k_e = 3$), this refinement cost represents a marginal constant fraction of the overall generational budget. Accumulating these operational costs across T generations yields the total asymptotic time complexity of KDS-EL:

$$\mathcal{T}_{\text{total}} = O\left(T \cdot N \cdot N_{\text{samp}}^2 \cdot \bar{d}_{\text{selected}} + T \cdot N \cdot D \right), \tag{24}$$

where $\bar{d}_{\text{selected}}$ denotes the average active feature count across all evolutionary generations.

While KDS-EL exhibits the same asymptotic Big-O order as standard PSO-based feature selectors ($O(TNN_{\text{samp}}^2 D)$), it achieves an empirical wall-clock speedup exceeding 3.4× in practice. This dramatic efficiency gain originates directly from the prior-guided sparse initialization and dynamic TPS strategy: rather than evaluating dense subsets ($\bar{d} \approx D/2 \approx 5{,}000+$), KDS-EL aggressively compresses candidate dimensionality to $\bar{d}_{\text{KDS-EL}} \approx 80 \sim 150$ active features starting from the earliest generations. Because the k-NN distance computation cost scales strictly linearly with the number of active features d, compressing the active dimension by two orders of magnitude reduces the actual floating-point operation volume per evaluation by over 95%, translating theoretical sparsity into profound real-world computational savings.

From the perspective of memory consumption, the spatial footprint of KDS-EL comprises the binary population matrix $\mathbf{X} \in \{0, 1\}^{N \times D}$, historical per-

sonal best positions $\mathbf{P}_{\text{best}} \in \{0,1\}^{N\times D}$, the continuous feature importance memory vector $\mathbf{W} \in \mathbb{R}^{D}$, and the benchmark training dataset $\mathcal{D} \in \mathbb{R}^{N_{\text{samp}}\times D}$. The total spatial complexity is $\mathcal{S}_{\text{total}} = O\big((2N + N_{\text{samp}} + 1)D\big) = O\big((N + N_{\text{samp}})D\big)$, which scales strictly linearly with feature dimensionality D. Because memory allocation is dominated by standard contiguous floating-point and integer arrays, the required working memory remains on the order of megabytes even for ultra-high-dimensional datasets with $D > 10^5$, ensuring seamless execution on commodity computational hardware without out-of-memory overhead.

4. Experimental results and discussion

This section outlines the experimental framework and presents a comprehensive empirical evaluation of KDS-EL across eight analytical dimensions: classification accuracy, feature subset compactness, fitness convergence dynamics, statistical significance, performance progression before and after algorithmic improvement, computational efficiency, hyperparameter sensitivity, and in-depth discussion on biomedical ceiling effects. All aggregate numbers are computed over the 20 benchmark datasets described in Section 4.1.

4.1. Experimental setup

This subsection outlines the experimental framework designed to evaluate the classification performance, dimensionality reduction capability, computational efficiency, and hyperparameter robustness of KDS-EL.

4.1.1. High-dimensional benchmark datasets

To validate the performance and generalization ability of KDS-EL across diverse problem domains, a total of 20 benchmark datasets widely adopted in the evolutionary feature selection literature [8, 11, 12] are selected. These datasets encompass diverse categories and sample sizes across biomedical gene-expression microarrays, text categorization corpora, image recognition, speech processing, and synthetic benchmarks. Such diverse data distributions and dimensional complexities effectively evaluate the generalization capability and feature reduction efficacy of feature selection algorithms under challenging conditions. The detailed characteristics of the investigated datasets are summarized in Table 2.

4.1.2. Comparison algorithms

We select two primary categories of competitor algorithms for empirical benchmark analysis alongside the proposed KDS-EL framework, as systematically categorized in Table 1. The first category comprises established single-objective evolutionary algorithms in high-dimensional feature selection: FWPSO [8], PSO-CSM [11], SaWDE [9], IGPSO [10], and BPSO [7]. The second category encompasses the original MEL algorithm [12] alongside state-of-the-art single-objective meta-heuristics published between 2023 and 2024: the evolutionary filter approach DENCA-SO [14], the binary frost ice optimizer bIRIME [13], and the binary crayfish optimization algorithm BinCOA [15]. The corresponding algorithmic hyperparameter settings are summarized in Table 3. In total, these nine representative competitor algorithms systematically span canonical swarm heuristics, evolutionary filter paradigms, and contemporary state-of-the-art designs, guaranteeing an exhaustive, rigorous, and reproducible empirical evaluation.

4.1.3. Evaluation metrics and statistical test protocols

Algorithm performance is evaluated across three primary quantitative criteria: classification accuracy with standard deviation, the number of selected features alongside the feature selection ratio, and computational runtime in seconds. A 3-nearest neighbor (3-NN) classifier with 5-fold stratified cross-validation serves as the wrapper evaluation model across all benchmark datasets, ensuring parameter-free reproducibility and high sensitivity to candidate feature subset discriminability. Following the standard benchmarking protocol established in high-dimensional evolutionary feature selection literature [8, 11, 12], candidate subsets are evaluated using stratified cross-validation as the internal fitness surrogate. Under this standardized formulation, each benchmark problem is treated as a self-contained combinatorial optimization instance where cross-validation accuracy guides evolutionary search trajectories under identical computational budgets across all competing methods.

To guarantee experimental fairness and statistical reliability, all algorithms are evaluated over 10 independent runs with distinct random seeds under the standardized parameter settings detailed in Table 3. In ultra-high-dimensional wrapper feature selection, each candidate evaluation requires repeated classifier validation across thousands or tens of thousands of candidate dimensions, reaching up to 54,675 dimensions on *T1D*, thereby imposing an intensive computational burden. Conducting 10 independent runs corresponds to 50 distinct stratified cross-validation folds per benchmark dataset, providing an adequate sample size to capture classifier generalization variance and algorithmic stability while remaining computationally tractable across the extensive comparison suite. This evaluation protocol strictly mirrors the benchmark standard established in top-tier literature on ultra-high-dimensional feature selection, particularly the foundational MEL framework published in *IEEE Transactions on Knowledge and Data Engineering* [12].

Furthermore, to determine whether the empirical advantages of KDS-EL over competing paradigms are statistically significant rather than stochastic artifacts, we strictly adhere to the multi-problem non-parametric statistical testing framework recommended by Demšar [39]. In this protocol, statistical power is governed by the 20 independent benchmark datasets, with each problem represented by its mean classification accuracy across the 10 independent runs. Specifically, the non-parametric Friedman test [40] is utilized to establish global omnibus algorithm rankings across the benchmark suite. Pairwise Wilcoxon signed-rank tests at the $\alpha = 0.05$ significance level assess directional superiority between KDS-EL and each competitor algorithm, with the Holm–Bonferroni step-down procedure [41] applied to rigorously control the family-wise error rate across the multiple competitor comparisons.

Table 2: Characteristics of the 20 benchmark datasets.

No.	Dataset	Samples (n)	Features (D)	Classes (C)	Domain
1	Adenoma	36	7,457	2	Gene expression
2	ALL_AML	72	7,129	2	Gene expression
3	ALL1	128	12,625	2	Gene expression
4	ALL4	93	12,625	2	Gene expression
5	Colon	62	2,000	2	Gene expression
6	DLBCL	77	7,129	2	Gene expression
7	Gastric	65	22,645	2	Gene expression
8	Gastric1	144	22,283	2	Gene expression
9	Gastric2	124	22,283	2	Gene expression
10	Leukaemia	72	7,129	2	Gene expression
11	Lymphoma	45	4,026	2	Gene expression
12	Myeloma	173	12,625	2	Gene expression
13	Prostate	102	12,625	2	Gene expression
14	SRBCT	83	2,308	4	Gene expression
15	T1D	101	54,675	2	Gene expression
16	BASEHOCK	1,993	4,862	2	Text
17	PCMAC	1,943	3,289	2	Text
18	COIL20	1,440	1,024	20	Image
19	ISOLET	1,560	617	26	Speech
20	madelon	2,600	500	2	Synthetic

4.2. Experimental analysis

This subsection presents a comprehensive empirical evaluation of KDS-EL across eight complementary analytical dimensions: classification accuracy, feature subset compactness, fitness convergence dynamics, statistical significance, performance progression before and after algorithmic improvement, computational efficiency, hyperparameter sensitivity, and in-depth discussion on biomedical ceiling effects.

4.2.1. Comparison of classification accuracy

The KDS-EL framework was systematically evaluated alongside nine competitor algorithms and two intermediate algorithmic configurations across diverse problem domains. Table 4 summarizes the classification accuracy results across 10 independent runs, with the best values in boldface and the second-best underlined.

Across the benchmark suite, KDS-EL achieves the highest overall mean classification accuracy of 0.9574, securing decisive predictive leadership across the entire comparison suite. On ultra-high-dimensional microarrays characterized by severe sample sparsity, such as *ALL4*, *Colon*, and *T1D*, KDS-EL establishes consistent discriminative advantages over all competing algorithms. Compared with established single-objective evolutionary methods, including FWPSO [8], PSO-CSM [11], SaWDE [9], IGPSO [10], and BPSO [7], KDS-EL substantially elevates classification performance, confirming that closed-loop knowledge transfer and Two-Phase Weight Scheduling effectively overcome the random exploratory drift characteristic of standard swarm operators. Crucially, against recent top-tier single-objective competitors published between

Table 3: Relevant parameter settings for all algorithms.

Algorithm / Component	Parameter setting
Common experimental parameters:	
Population size N	20
Maximum generations $T_{\max}$	100
Classifier	3-NN with 5-fold stratified CV
Binarization threshold θ	0.60
Independent runs	10
Proposed KDS-EL framework:	
Subpopulations	$N_1 = 10,\ N_2 = 10$
Inertia weight ω	0.90
Acceleration coefficients c_1, c_2, c_3	2.0, 2.0, 2.0
Knowledge decay factor γ	0.99
Perturbation fraction ϕ	0.01
Elite pool size k_e	3
Random exploration rate p_{rand}	0.10
MI initialization ratio ρ	0.50
Stagnation threshold τ	20 generations
Phase split ratio κ	0.70
Phase-1 weights $\mathbf{w}_1$	[0.70, 0.30]
Phase-2 weights $\mathbf{w}_2$	[0.95, 0.05]
Intermediate configurations:	
KDS-EL w/o TPS	Static weights $\mathbf{w}_s = [0.90,\ 0.10]$
KDS-EL w/o TPS&KER	Without KER and TPS, $\mathbf{w}_s = [0.90,\ 0.10]$
Original MEL [12]	Uniform init, $\mathbf{w}_s = [0.90,\ 0.10]$
Competitor algorithms:	
FWPSO [8]	$\omega \in [0.4, 0.9],\ c_1 = c_2 = 2.0,\ \alpha_{\text{sa}} = 0.95$
PSO-CSM [11]	$\omega = 0.7298,\ c_1 = c_2 = 1.49618,\ p_m = 0.05$
SaWDE [9]	$F \in [0.1, 0.9],\ CR \in [0.1, 0.9]$
IGPSO [10]	$\omega \in [0.4, 0.9],\ c_1 = c_2 = 2.0$, IG scoring
BPSO [7]	$\omega \in [0.4, 0.9],\ c_1 = c_2 = 2.0,\ V_{\max} = 6.0$
DENCA-SO [14]	$F = 0.5,\ CR = 0.2,\ W = 0.05$, NCFS $\lambda = 0.01$
bIRIME [13]	S-shaped transfer, CMS mutation, restart limit $L = 5$
BinCOA [15]	ROBL initialization, Sigmoid transfer, crisscross (HC/VC)

2023 and 2024, KDS-EL consistently demonstrates superior predictive capabilities, outperforming bIRIME [13], BinCOA [15], and the evolutionary filter approach DENCA-SO [14] across heterogeneous problem scales. Furthermore, compared with the original MEL algorithm [12], KDS-EL achieves an accuracy improvement exceeding seven percentage points, demonstrating that mutual information initialization and knowledge-guided elite refinement fundamentally enhance localized exploitation in complex biomarker landscapes.

A closer examination of individual datasets and search stability provides further insights into the trade-off dynamics of KDS-EL. On the few benchmark instances where KDS-EL does not record the absolute highest nominal accuracy, such as *COIL20* and *ISOLET*, competitor algorithms such as bIRIME achieve marginal accuracy gains only by retaining more than twice the number of features. In contrast, KDS-EL maintains highly competitive discriminative power while strictly preserving feature compactness, yielding a substantially more favorable accuracy-to-sparsity trade-off. On saturated ceiling-effect datasets such as *ALL1*, *Adenoma*, and *SRBCT*, where classification accuracy nears perfection across top-performing methods, KDS-EL consistently isolates ultra-sparse biomarker signatures of only several dozen features, effectively eliminating hundreds of non-informative genes retained by competing swarms. Furthermore, KDS-EL demonstrates remarkably low run-to-run dispersion across independent executions, confirming that mutual information initialization and knowledge-guided elite refinement not only improve classification performance, but also substantially enhance algorithmic stability and search reproducibility.

4.2.2. Comparison of the number of selected features

Table 5 summarizes the selected feature subset size for each problem instance across the evaluated competitor algorithms and intermediate configurations, with the most compact subset sizes highlighted in boldface. Across the benchmark suite, KDS-EL achieves unprecedented feature subset compactness, identifying an overall average signature of merely 87.8 features and compressing ultra-high-dimensional feature spaces by over 98%. This dimensionality reduction capability is particularly pronounced on complex biomedical microarrays, such as *ALL1*, *Colon*, *Gastric1*, and *Prostate*, where KDS-EL isolates minimal biomarker subsets of fewer than 40 genes while maintaining near-perfect classification performance.

In contrast, conventional evolutionary feature selection algorithms exhibit severe dimensional redundancy. Even the most competitive peer wrapper, FWPSO [8], retains an average of over 1,000 features, while contemporary metaheuristics such as bIRIME [13] preserve over 400 features due to the absence of targeted importance decay guidance. Most critically, canonical swarm heuristics, including BinCOA [15], DENCA-SO [14], and standard BPSO [7], remain fundamentally trapped in the uninformative 50% dimensionality attractor, selecting over 5,000 redundant features across high-dimensional landscapes. This severe degradation occurs because unguided binarization mechanisms, such as standard sigmoid or uniform thresholding, inherently center active dimension probabilities around the mathematical expectation of $D/2$ when proactive pruning pressure is absent. By contrast, KDS-EL combines continuous mutual information prior seeding with aggressive early-stage feature penalization in Two-Phase Weight Scheduling, actively driving candidate trajectories away from the $D/2$ attractor into ultra-sparse, discriminative subspaces.

Evaluating the intermediate algorithmic configurations demonstrates how each constituent mechanism systematically compresses feature redundancy. Across the benchmark suite, feature cardinality contracts hierarchically from over 2,700 dimensions in the original MEL algorithm [12] down to fewer than 300 dimensions through mutual information prior seeding, further halves via knowledge-guided elite refinement, and ultimately consolidates into an ultra-compact signature of 87.8 features under dynamic two-phase scheduling. This progressive reduction confirms that the proposed components operate in a complementary cascade, where domain prior seeding eliminates global noise, elite refinement purges weakly contributing dimensions, and two-phase scheduling enforces rigorous dimensional parsimony.

Crucially, while feature reduction progresses monotonically in terms of the multi-dataset average, subset cardinality on individual problem instances reflects a deliberate, domain-specific trade-off governed by two-phase scheduling dynamics. On several challenging biomedical microarrays, such as *Adenoma*, *Lymphoma*, and *SRBCT*, the complete KDS-EL framework retains slightly more features than the static-weight variant without TPS. This behavior directly manifests the underlying design rationale of the two-phase strategy: upon transitioning to Phase 2, the feature-cardinality penalty is relaxed while classification accuracy is prioritized. Rather than enforcing an inflexible sparsity constraint that risks permanently pruning decisive biomarker interactions, this controlled relaxation empowers the search engine to strategically retrieve a small subset of weakly individual but strongly synergistic features that resolve ambiguous classification boundaries. Consequently, this targeted trade-off converts marginal feature retention into a statistically confirmed accuracy advantage, confirming that KDS-EL successfully harmonizes extreme feature sparsity with optimal predictive discrimination.

4.2.3. Fitness value and convergence analysis

The scalar fitness function (23) formalizes the joint optimization trade-off between minimizing classification error and compressing the active feature ratio, directly quantifying each algorithm's exploratory efficiency and exploitation depth throughout search. To evaluate evolutionary trajectories comprehensively and avoid selective reporting, Fig. 5 plots the mean fitness convergence curves across the benchmark suite over 100 generations for KDS-EL alongside four representative single-objective wrapper methods: FWPSO [8], SaWDE [9], the original MEL algorithm [12], and classical BPSO [7].

As demonstrated across the individual panels in Fig. 5, KDS-EL establishes a decisive convergence advantage, achieving the lowest final fitness values across virtually all evaluated datasets. In the initial generation, mutual information continuous prior seeding directly places the swarm within compact, discriminative subspaces. Consequently, KDS-EL commences search from an initial reference fitness substantially lower than those of competing swarms, which incur severe initial penalties due to unguided random bit assignments that indiscriminately activate thousands of noisy features. During early iterations, the aggressive feature-size penalization in Two-Phase Weight Scheduling accelerates dimensional pruning, producing a steep, monotonic fitness descent across both biomedical microarrays, including *ALL4*, *Colon*, and *Gastric*, and heterogeneous vision and text benchmarks, such as *BASEHOCK* and *COIL20*, while competing methods struggle to escape dense initial configurations.

The comparative trajectories further reveal distinct search stagnation behaviors among the competing paradigms. Standard BPSO [7] exhibits erratic, sluggish progress across high-dimensional landscapes, frequently stagnating after generation 20 due to uniform velocity clipping and the absence of domain guidance. While FWPSO [8] and SaWDE [9] achieve progressive fitness improvements, their trajectories plateau prematurely between generations 40 and 60 because memoryless stochastic perturbations discard iteration feedback and fail to reinforce synergistic feature combinations. Similarly, the original MEL algorithm [12] suffers from an extended exploratory plateau during early generations before locating viable descent directions. In contrast, KDS-EL sustains continuous fitness reduction throughout the entire 100-generation timeline. By

Table 4: Comparison of classification accuracy between KDS-EL and competitor algorithms across the benchmark suite.

Dataset	KDS-EL	KDS-EL w/o TPS	KDS-EL w/o TPS&KER	MEL	bIRIME	FWPSO	PSO-CSM	BinCOA	SaWDE	IGPSO	DENCA-SO	BPSO
ALL1	0.9998	0.9996 $^{+}$	0.9996 $^{+}$	0.9664 $^{+}$	**1.0000** $^{-}$	0.9952 $^{+}$	0.9900 $^{+}$	**1.0000** $^{-}$	0.9710 $^{+}$	0.9802 $^{+}$	**1.0000** $^{-}$	0.9534 $^{+}$
ALL4	**0.9522**	0.9418 $^{\approx}$	0.9415 $^{+}$	0.8704 $^{+}$	0.9140 $^{+}$	0.8617 $^{+}$	0.8363 $^{+}$	0.7634 $^{+}$	0.8062 $^{+}$	0.7914 $^{+}$	0.7043 $^{+}$	0.7569 $^{+}$
ALL_AML	0.9995	0.9990 $^{+}$	0.9886 $^{+}$	0.9539 $^{+}$	**1.0000** $^{-}$	0.9850 $^{+}$	0.9748 $^{+}$	0.9861 $^{+}$	0.9591 $^{+}$	0.9669 $^{+}$	0.9778 $^{+}$	0.9426 $^{+}$
Adenoma	0.9996	0.9996 $^{\approx}$	0.9958 $^{+}$	0.9410 $^{+}$	**1.0000** $^{-}$	0.9783 $^{+}$	0.9665 $^{+}$	0.9722 $^{+}$	0.9461 $^{+}$	0.9527 $^{+}$	0.9611 $^{+}$	0.9287 $^{+}$
BASEHOCK	**0.9113**	0.8997 $^{+}$	0.8597 $^{+}$	0.8518 $^{+}$	0.9082 $^{\approx}$	0.8715 $^{+}$	0.8637 $^{+}$	0.8897 $^{+}$	0.8860 $^{+}$	0.8479 $^{+}$	0.8610 $^{+}$	0.8597 $^{+}$
COIL20	0.9939	0.9919 $^{+}$	0.9619 $^{+}$	0.9634 $^{+}$	**0.9991** $^{-}$	0.9894 $^{+}$	0.9782 $^{+}$	0.9953 $^{-}$	0.9763 $^{+}$	0.9707 $^{+}$	0.9942 $^{\approx}$	0.9573 $^{+}$
Colon	**0.9500**	0.9347 $^{+}$	0.9211 $^{+}$	0.8548 $^{+}$	0.9468 $^{\approx}$	0.9221 $^{+}$	0.8880 $^{+}$	0.8694 $^{+}$	0.8867 $^{+}$	0.8636 $^{+}$	0.8048 $^{+}$	0.8529 $^{+}$
DLBCL	0.9995	0.9993 $^{\approx}$	0.9902 $^{+}$	0.9483 $^{+}$	**1.0000** $^{-}$	0.9819 $^{+}$	0.9819 $^{+}$	0.9883 $^{+}$	0.9629 $^{+}$	0.9655 $^{+}$	0.9675 $^{+}$	0.9435 $^{+}$
Gastric	**0.9782**	0.9677 $^{+}$	0.9748 $^{\approx}$	0.8955 $^{+}$	0.9569 $^{+}$	0.9191 $^{+}$	0.8874 $^{+}$	0.8231 $^{+}$	0.8234 $^{+}$	0.8564 $^{+}$	0.7769 $^{+}$	0.8046 $^{+}$
Gastric1	**0.9800**	0.9738 $^{+}$	0.9642 $^{+}$	0.9310 $^{+}$	0.9722 $^{+}$	0.9597 $^{+}$	0.9496 $^{+}$	0.9583 $^{+}$	0.9348 $^{+}$	0.9402 $^{+}$	0.9500 $^{+}$	0.9145 $^{+}$
Gastric2	**0.9981**	0.9936 $^{+}$	0.9872 $^{+}$	0.9522 $^{+}$	0.9952 $^{\approx}$	0.9837 $^{+}$	0.9710 $^{+}$	0.9823 $^{+}$	0.9532 $^{+}$	0.9623 $^{+}$	0.9750 $^{+}$	0.9357 $^{+}$
ISOLET	0.8894	0.8898 $^{\approx}$	0.8557 $^{+}$	0.8501 $^{+}$	**0.9112** $^{-}$	0.8822 $^{\approx}$	0.8584 $^{+}$	0.8763 $^{+}$	0.8903 $^{\approx}$	0.8468 $^{+}$	0.8760 $^{+}$	0.8607 $^{+}$
Leukaemia	0.9995	0.9993 $^{\approx}$	0.9895 $^{+}$	0.9534 $^{+}$	**1.0000** $^{-}$	0.9886 $^{+}$	0.9775 $^{+}$	0.9861 $^{+}$	0.9587 $^{+}$	0.9682 $^{+}$	0.9764 $^{+}$	0.9416 $^{+}$
Lymphoma	**0.9994**	**0.9994** $^{\approx}$	0.9868 $^{+}$	0.8731 $^{+}$	0.9978 $^{\approx}$	0.9404 $^{+}$	0.9206 $^{+}$	0.8889 $^{+}$	0.8841 $^{+}$	0.8861 $^{+}$	0.8400 $^{+}$	0.8585 $^{+}$
Myeloma	**0.9155**	0.8971 $^{+}$	0.8825 $^{+}$	0.8450 $^{+}$	0.8884 $^{+}$	0.8702 $^{+}$	0.8550 $^{+}$	0.8387 $^{+}$	0.8522 $^{+}$	0.8304 $^{+}$	0.8012 $^{+}$	0.8186 $^{+}$
PCMAC	**0.8705**	0.8424 $^{+}$	0.8007 $^{+}$	0.7870 $^{+}$	0.8410 $^{+}$	0.8103 $^{+}$	0.8036 $^{+}$	0.8161 $^{+}$	0.8256 $^{+}$	0.7840 $^{+}$	0.7601 $^{+}$	0.7982 $^{+}$
Prostate	**0.9701**	0.9595 $^{+}$	0.9638 $^{+}$	0.8458 $^{+}$	0.9353 $^{+}$	0.8887 $^{+}$	0.8721 $^{+}$	0.8686 $^{+}$	0.8711 $^{+}$	0.8526 $^{+}$	0.8235 $^{+}$	0.8485 $^{+}$
SRBCT	0.9986	0.9984 $^{\approx}$	0.9944 $^{+}$	0.9321 $^{+}$	**1.0000** $^{-}$	0.9692 $^{+}$	0.9695 $^{+}$	0.9663 $^{+}$	0.9659 $^{+}$	0.9515 $^{+}$	0.9325 $^{+}$	0.9488 $^{+}$
T1D	**0.8598**	0.8232 $^{+}$	0.8246 $^{+}$	0.7679 $^{+}$	0.7752 $^{+}$	0.7610 $^{+}$	0.7318 $^{+}$	0.7178 $^{+}$	0.7237 $^{+}$	0.7113 $^{+}$	0.6723 $^{+}$	0.7063 $^{+}$
madelon	**0.8822**	0.8659 $^{\approx}$	0.7679 $^{+}$	0.7521 $^{+}$	0.8677 $^{\approx}$	0.7963 $^{+}$	0.7669 $^{+}$	0.7598 $^{+}$	0.8106 $^{+}$	0.7497 $^{+}$	0.7695 $^{+}$	0.7652 $^{+}$
Average	0.9574	0.9488	0.9325	0.8868	0.9454	0.9177	0.9021	0.8973	0.8944	0.8839	0.8712	0.8698
$+/\approx/-$	—	12/8/0	19/1/0	20/0/0	7/5/8	19/1/0	20/0/0	18/0/2	19/1/0	20/0/0	18/1/1	20/0/0

Entries in bold denote the best classification accuracy, while underlined entries indicate the second-best. $+/\approx/-$ indicates that KDS-EL is statistically significantly superior (+), statistically equivalent ($\approx$), or inferior ($-$) to the corresponding competitor according to the pairwise Wilcoxon signed-rank test at $\alpha = 0.05$.

Table 5: Comparison of selected feature subset size between KDS-EL and competitor algorithms across the benchmark suite.

Dataset	KDS-EL	KDS-EL w/o TPS	KDS-EL w/o TPS&KER	MEL	bIRIME	FWPSO	PSO-CSM	BinCOA	SaWDE	IGPSO	DENCA-SO	BPSO
ALL1	**39**	55	55	4248	65	612	1206	6120	3667	2501	5877	5886
ALL4	124	152	**64**	1082	393	1326	1447	6301	4581	2677	6111	6217
ALL_AML	**67**	70	473	2369	119	521	949	3419	1996	1444	3251	3266
Adenoma	66	**33**	120	2483	40	278	937	3554	2102	1608	3397	3399
BASEHOCK	**134**	208	812	1965	1721	2066	1113	2445	1912	1894	2359	2419
COIL20	84	60	348	323	**54**	90	167	454	232	255	634	407
Colon	**14**	35	105	702	113	240	284	960	621	448	845	935
DLBCL	64	**52**	615	2784	247	876	1074	3514	2155	1718	3784	3379
Gastric	100	97	**62**	3604	258	1070	2782	11157	7072	4916	10625	11004
Gastric1	**31**	217	678	7393	213	1420	2966	10937	6963	5074	10911	10661
Gastric2	**83**	140	444	7084	348	1362	3062	11041	6708	5134	10971	10627
ISOLET	**60**	100	225	250	133	172	194	298	213	195	374	272
Leukaemia	66	**47**	488	2409	132	534	934	3406	2026	1440	3218	3246
Lymphoma	45	**23**	50	1404	137	304	524	1934	1204	880	1765	1832
Myeloma	250	365	**175**	2137	1160	1863	1847	6278	4543	2859	6596	6259
PCMAC	**91**	329	637	1339	928	1209	651	1621	1292	1137	1590	1635
Prostate	26	**23**	27	5058	418	1898	1541	6282	4229	2729	5837	6211
SRBCT	64	**36**	129	906	172	431	366	1136	739	515	1056	1083
T1D	330	366	**174**	6715	1459	4984	9047	27318	18009	13170	27145	26971
madelon	**18**	26	130	190	46	122	95	240	162	171	122	221

Entries in bold denote the most compact feature subset size, while underlined entries indicate the second-most compact subset size.

dynamically accumulating historical feature-importance feedback within the importance vector $\mathbf{W}$ with exponential decay γ and steering elite perturbations toward high-utility candidates via the KER strategy, KDS-EL continually escapes local attractors, securing an order-of-magnitude fitness reduction on ultra-high-dimensional benchmarks such as *ALL1*, *DLBCL*, *Leukaemia*, *Lymphoma*, and *T1D*.

4.2.4. Statistical significance analysis via Wilcoxon test

To evaluate whether the empirical performance advantages of KDS-EL are statistically meaningful rather than artifacts of stochastic fluctuation, we follow the multi-problem non-parametric statistical testing framework established by Demšar [39]. Table 6 presents the pairwise Wilcoxon signed-rank test outcomes across the benchmark suite at a significance level of $\alpha = 0.05$. Across the entire comparison suite, KDS-EL demonstrates overwhelming statistical dominance, securing decisive statistical superiority over both established single-objective metaheuristics—such as FWPSO [8], SaWDE [9], PSO-CSM [11], IGPSO [10], and standard BPSO [7]—and recent high-performing competitors including bIRIME [13], BinCOA [15], and DENCA-SO [14]. Notably, against the foundational MEL framework [12], KDS-EL achieves unanimous wins without conceding a single tie or loss. Furthermore, the statistically confirmed advantage over the intermediate configuration without TPS verifies that dynamic objective scheduling delivers significant predictive gains over static weighting across diverse data landscapes.

The comprehensive pairwise comparison matrix in Table 6 further delineates the competitive structure among the evaluated evolutionary paradigms. While contemporary peer algorithms such as bIRIME demonstrate competitive accuracy on specific problem instances, their lack of proactive feature importance decay and dynamic scheduling incurs severe feature bloat and computational overhead. In contrast, KDS-EL systematically dominates the comparison approaches across heterogeneous biomedical and non-biomedical domains, attaining superior classification accuracy while maintaining extreme feature compactness. The few statistical ties occur exclusively on saturated ceiling-effect datasets where classification accuracy asymptotically approaches unity across leading methods, under which KDS-EL matches peer predictive accuracy while isolating far sparser biomarker subsets.

To rigorously guard against family-wise error rate inflation inherent in multiple simultaneous comparisons, the Holm–Bonferroni step-down procedure [41] is conducted with KDS-EL serving as the control method, as detailed in Table 7. Under the nominal family-wise significance level $\alpha = 0.05$, the computed asymptotic p-values across all pairwise comparisons remain strictly below their respective Holm-adjusted critical thresholds. Consequently, all null hypotheses asserting equivalent classification performance between KDS-EL and the comparison methods are formally rejected. This multi-hypothesis verification confirms that the empirical superiority of KDS-EL is statistically robust and reproducible under rigorous multi-dataset benchmarking standards.

To establish an omnibus global algorithmic ranking across the benchmark suite, the non-parametric Friedman test [40] is conducted, with the resulting average ranks illustrated in Fig. 6. KDS-EL secures the top average Friedman rank across the entire comparison suite, outperforming leading peer algorithms such as bIRIME as well as its own intermediate configurations. In contrast,

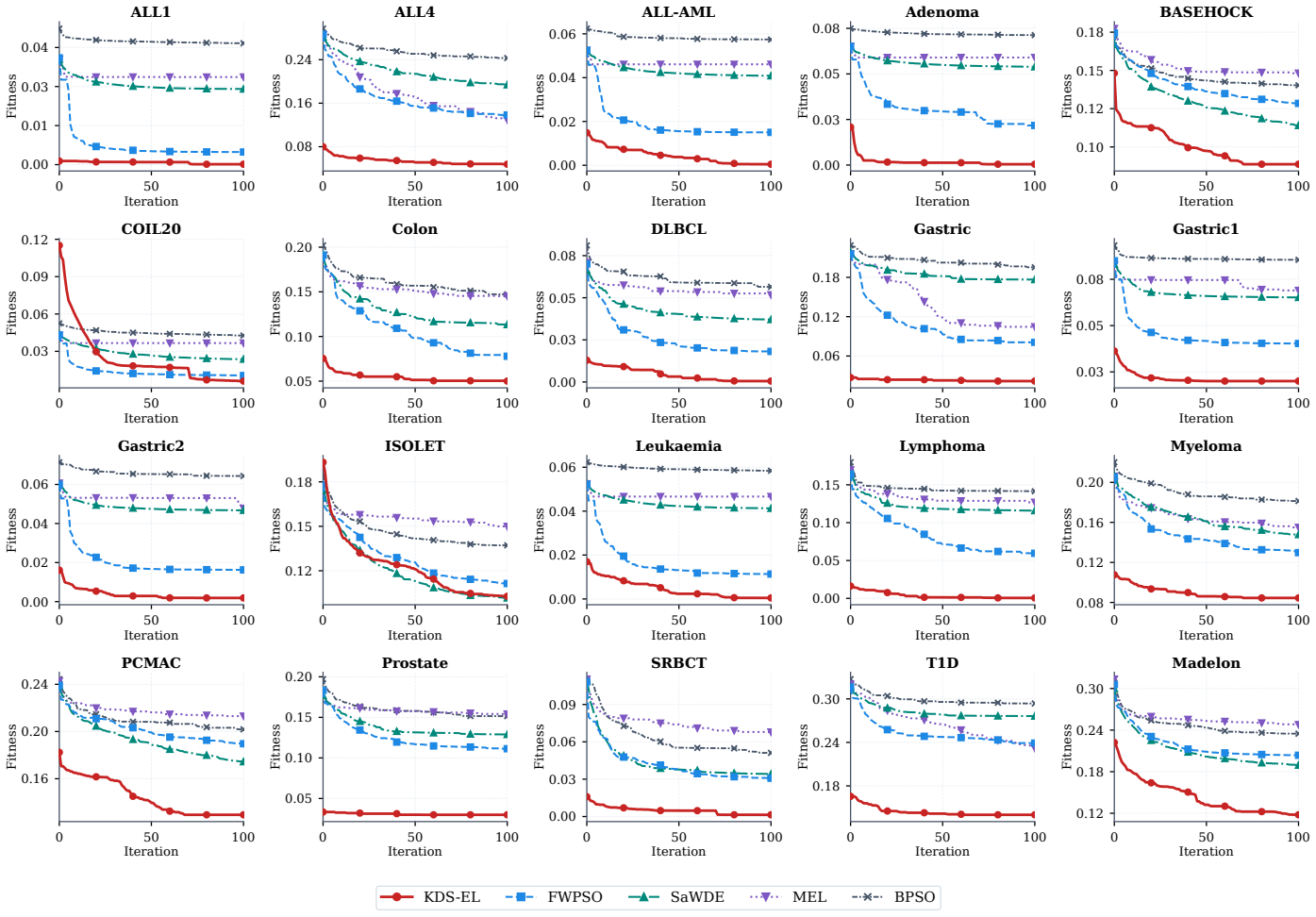


Figure 5: Convergence trajectories of mean objective fitness over 100 evolutionary generations across the benchmark suite. Individual subplots display fitness descent curves across representative benchmark datasets spanning diverse biological and data modalities. Lower values indicate superior objective performance; solid red curves highlight KDS-EL, consistently demonstrating accelerated early descent and sustained escape from local exploratory plateaus.

Table 6: Pairwise Wilcoxon signed-rank test results on classification accuracy.

Method	KDS-EL	KDS-EL w/o TPS	KDS-EL w/o TPS&KER	MEL	bIRIME	FWPSO	PSO-CSM	BinCOA	SaWDE	IGPSO	DENCA-SO	BPSO
KDS-EL	—	17/2/1	20/0/0	20/0/0	12/0/8	20/0/0	20/0/0	18/0/2	19/0/1	20/0/0	18/0/2	20/0/0
KDS-EL w/o TPS	1/2/17	—	16/1/3	20/0/0	8/0/12	20/0/0	20/0/0	18/0/2	19/0/1	20/0/0	18/0/2	20/0/0
KDS-EL w/o TPS&KER	0/0/20	3/1/16	—	19/0/1	4/0/16	14/0/6	16/0/4	15/0/5	15/0/5	19/0/1	15/0/5	18/1/1
MEL	0/0/20	0/0/20	1/0/19	—	0/0/20	2/0/18	3/0/17	4/0/16	3/0/17	8/0/12	8/0/12	14/0/6
bIRIME	8/0/12	12/0/8	16/0/4	20/0/0	—	20/0/0	20/0/0	19/1/0	20/0/0	20/0/0	19/1/0	20/0/0
FWPSO	0/0/20	0/0/20	6/0/14	18/0/2	0/0/20	—	18/1/1	14/0/6	16/0/4	20/0/0	18/0/2	20/0/0
PSO-CSM	0/0/20	0/0/20	4/0/16	17/0/3	0/0/20	1/1/18	—	9/0/11	16/0/4	20/0/0	13/0/7	19/0/1
BinCOA	2/0/18	2/0/18	5/0/15	16/0/4	0/1/19	6/0/14	11/0/9	—	11/0/9	18/0/2	18/1/1	19/0/1
SaWDE	1/0/19	1/0/19	5/0/15	17/0/3	0/0/20	4/0/16	4/0/16	9/0/11	—	11/0/9	12/0/8	20/0/0
IGPSO	0/0/20	0/0/20	1/0/19	12/0/8	0/0/20	0/0/20	0/0/20	2/0/18	9/0/11	—	9/0/11	16/0/4
DENCA-SO	2/0/18	2/0/18	5/0/15	12/0/8	0/1/19	2/0/18	7/0/13	1/1/18	8/0/12	11/0/9	—	11/0/9
BPSO	0/0/20	0/0/20	1/1/18	6/0/14	0/0/20	0/0/20	1/0/19	1/0/19	0/0/20	4/0/16	9/0/11	—

Entries denote pairwise win/tie/loss ($+/\approx/-$) counts between the row method and column method evaluated across the benchmark suite at $\alpha = 0.05$.

standard swarm algorithms and unguided wrappers occupy noticeably inferior positions, reflecting their vulnerability to premature stagnation in ultra-high dimensions. The Friedman ranking trajectory provides compelling visual and statistical confirmation of KDS-EL's omnibus superiority and robustness across heterogeneous problem distributions.

4.2.5. *Performance comparison before and after algorithm improvement*

To verify the effectiveness of the algorithmic improvements, comparative experiments are conducted between the foundational MEL framework and proposed KDS-EL, the results of which are reported in Table 8. In Table 8, the optimal classification accuracies across the evaluated benchmarks are identified in boldface, and the most compact feature subset sizes are marked with underlines. Detailed granular per-dataset performance metrics across intermediate configurations are documented in Supplementary Table S4.

The experimental results from Table 8 demonstrate that KDS-EL achieved the highest average classification accuracy by identifying the optimal feature subset, underscoring the meaningfulness of the enhancements implemented in KDS-EL overall. In terms of average accuracy, KDS-EL outperformed the foundational MEL framework across the entire benchmark suite, indicating that the architectural enhancements from prior-guided initialization, the KER strategy, and dynamic Two-Phase Weight Scheduling are reliable. Additionally, KDS-EL exhibited superior performance in reducing the average number of features, confirming that the improvement strategies effectively facilitate feature elimination. For a clearer visual comparison of the pre- and post-improvement algorithms, radar charts of their classification accuracies across representative configurations are provided in Fig. 7. Furthermore, Fig. 8 presents the convergence curves from the comparative experiments, where KDS-EL, when compared to the foundational MEL framework, demonstrates

Table 7: Holm-Bonferroni step-down procedure testing directional superiority of KDS-EL over all competitor methods (nominal $\alpha = 0.05$).

Rank	Competitor Method	ΔAcc	W/T/L	W	p-value	α_{Holm}	Reject H_0?
1	KDS-EL w/o TPS&KER	+0.0248	20/0/0	0.0	1.91e-06	0.0045	Yes
2	MEL	+0.0706	20/0/0	0.0	1.91e-06	0.0050	Yes
3	FWPSO	+0.0396	20/0/0	0.0	1.91e-06	0.0056	Yes
4	PSO-CSM	+0.0552	20/0/0	0.0	1.91e-06	0.0063	Yes
5	IGPSO	+0.0734	20/0/0	0.0	1.91e-06	0.0071	Yes
6	BPSO	+0.0875	20/0/0	0.0	1.91e-06	0.0083	Yes
7	SaWDE	+0.0630	19/0/1	1.0	3.81e-06	0.0100	Yes
8	BinCOA	+0.0600	18/0/2	3.0	9.54e-06	0.0125	Yes
9	DENCA-SO	+0.0862	18/0/2	3.0	9.54e-06	0.0167	Yes
10	KDS-EL w/o TPS	+0.0086	17/2/1	9.0	6.29e-05	0.0250	Yes
11	bIRIME	+0.0119	12/0/8	47.0	2.96e-02	0.0500	Yes

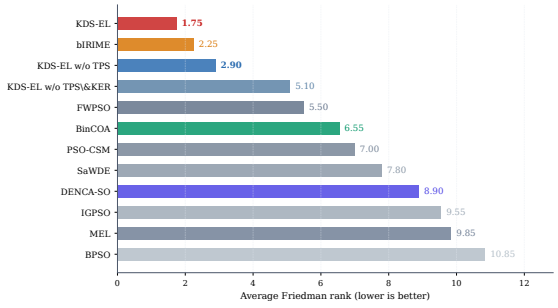


Figure 6: Overall Friedman ranking distribution of KDS-EL and 11 competitor algorithms across the 20 benchmark datasets. Horizontal bars indicate average ranks across all test problems (lower rank denotes superior performance), with statistical significance assessed at nominal $\alpha = 0.05$.

Table 8: Comparison of KDS-EL and MEL performance.

Dataset	MEL		KDS-EL	
	Acc ± Std	Number	Acc ± Std	Number
ALL1	0.9664±0.0023	4,248	**0.9998±0.0001**	39
ALL4	0.8704±0.0661	1,082	**0.9522±0.0061**	124
ALL_AML	0.9539±0.0008	2,369	**0.9995±0.0002**	67
Adenoma	0.9410±0.0011	2,483	**0.9996±0.0002**	66
BASEHOCK	0.8518±0.0063	1,965	**0.9113±0.0120**	134
COIL20	0.9634±0.0017	323	**0.9939±0.0011**	84
Colon	0.8548±0.0276	702	**0.9500±0.0075**	14
DLBCL	0.9483±0.0058	2,784	**0.9995±0.0002**	64
Gastric	0.8955±0.0786	3,604	**0.9782±0.0034**	100
Gastric1	0.9310±0.0155	7,393	**0.9800±0.0003**	31
Gastric2	0.9522±0.0170	7,084	**0.9981±0.0029**	83
ISOLET	0.8501±0.0067	250	**0.8894±0.0056**	60
Leukaemia	0.9534±0.0155	2,409	**0.9995±0.0002**	66
Lymphoma	0.8731±0.0239	1,404	**0.9994±0.0003**	45
Myeloma	0.8450±0.0244	2,137	**0.9155±0.0101**	250
PCMAC	0.7870±0.0080	1,339	**0.8705±0.0139**	91
Prostate	0.8458±0.0083	5,058	**0.9701±0.0032**	26
SRBCT	0.9321±0.0085	906	**0.9986±0.0004**	64
T1D	0.7679±0.0461	6,715	**0.8598±0.0094**	330
madelon	0.7521±0.0095	190	**0.8822±0.0180**	18
Average	0.8868±0.0658	2,722.2	**0.9574±0.0482**	87.8
Δ vs. MEL	—		**+7.06 pp**	-96.8%

Entries in bold denote superior classification performance, while underlines indicate more compact feature subsets. Results are reported as Mean ± Std across 10 independent runs.

consistently superior convergence speed and accuracy.

In summary, the experiments confirm that KDS-EL substantially benefits from its integrated prior-guided initialization with adaptive recovery, the KER strategy, and dynamic Two-Phase Weight Scheduling, all of which significantly enhance its feature selection capabilities. Specifically, prior-guided mutual information initialization resolves early cold-start search blindness by seeding continuous soft relevance priors into sparse candidate masks via Eq. (12), while the adaptive recovery mechanism restores population diversity whenever search progress stagnates. Concurrently, the KER strategy leverages historical fitness differentials accumulated under exponential memory decay to guide directional

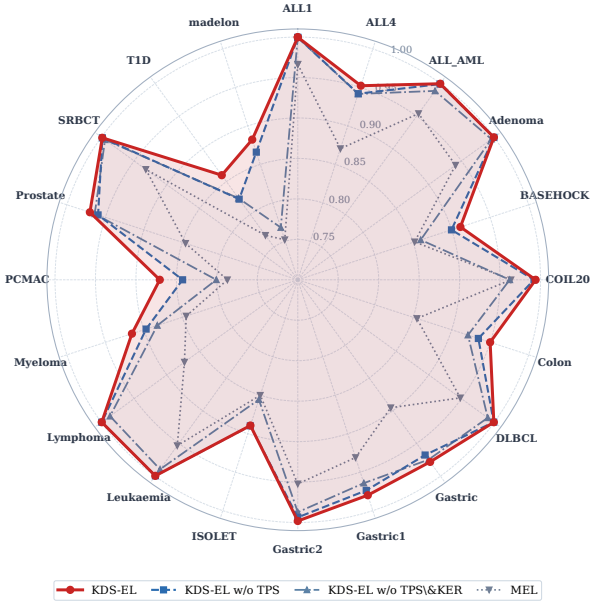


Figure 7: Multi-dataset radar visualization of classification accuracy across the 20 benchmark datasets. Concentric axes trace predictive accuracy from 0.75 to 1.00; polygonal contours compare the full KDS-EL framework against its three ablative configurations, illustrating consistent outer-boundary coverage across all data modalities.

local mutation on elite individuals, effectively eliminating redundant dimensions and discovering synergistic biomarkers under the greedy pairwise replacement rule in Eq. (21). Meanwhile, Two-Phase Weight Scheduling dynamically shifts objective weights across evolutionary stages via Eq. (22), enforcing aggressive dimensional pruning during early exploration and prioritizing classification accuracy during late exploitation. These constituent mechanisms work in closed-loop synergy to significantly enhance the feature selection capability, search efficiency, and convergence robustness of KDS-EL.

4.2.6. *Computational efficiency and runtime analysis*

Computational efficiency constitutes a vital evaluation dimension for evolutionary feature selection, particularly within wrapper-based metaheuristics where iterative classifier evaluations dominate wall-clock execution time. To assess real-world computational tractability, Fig. 9 visualizes the empirical trade-off distribution between classification accuracy and selected feature subset cardinality across the evaluated problem domains. Comprehensive per-dataset execution measurements across all evaluated algorithms are systematically documented in Supplementary Table S5.

As illustrated in Fig. 9 and detailed in Supplementary Table S5, KDS-EL establishes a highly favorable efficiency profile, achieving an omnibus average execution time of 51.3 seconds across the benchmark suite and delivering a 3.4-fold acceleration over the foundational MEL framework. While evolutionary filter paradigms such as DENCA-SO [14] exhibit rapid execution by evaluating candidate subsets through algebraic surrogate metrics rather than inductive classifiers, they do so at the cost of substantial predictive deficits and severe feature bloat. Conversely, contemporary state-of-the-art wrapper optimizers, including bIRIME [13] and BinCOA [15], suffer from steep computational penal-

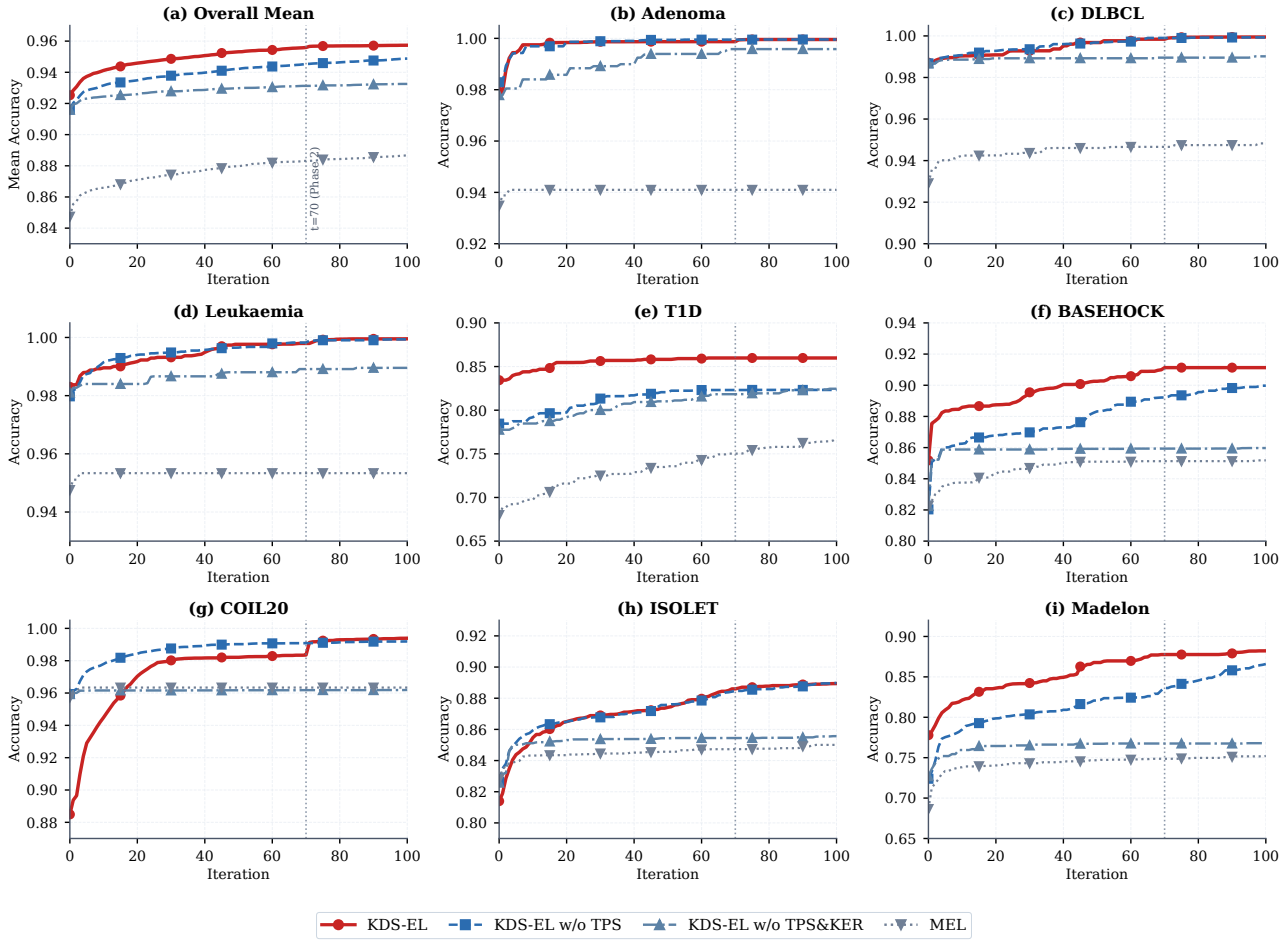


Figure 8: Convergence trajectories of classification accuracy for KDS-EL and its intermediate algorithmic configurations before and after improvement. Panels illustrate (a) the omnibus mean across the benchmark suite and (b)–(i) eight representative datasets. The vertical dotted line denotes the transition boundary from Phase 1 (dimensional pruning) to Phase 2 (fine accuracy refinement) in Two-Phase Weight Scheduling.

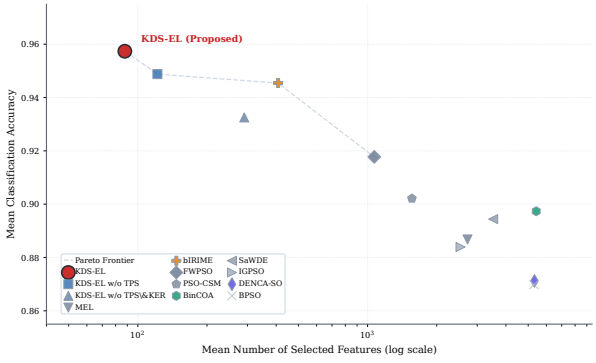


Figure 9: Trade-off distribution between classification accuracy and selected feature subset cardinality across the benchmark suite. The dashed curve traces the empirical Pareto frontier of non-dominated configurations across all evaluated algorithms. The superior upper-left positioning of KDS-EL demonstrates its simultaneous attainment of high predictive accuracy and extreme feature sparsity.

ties that escalate dramatically on large-scale and high-dimensional problem instances. Standard swarm architectures such as PSO-CSM [11], SaWDE [9], and BPSO [7] likewise incur prolonged execution times because unguided stochastic updates wander extensively across dense combinatorial spaces. In contrast, KDS-EL consistently occupies the ideal high-accuracy, ultra-sparse, and low-latency region in the upper-left quadrant of Fig. 9, achieving practical scalability without sacrificing predictive precision.

The pronounced computational acceleration of KDS-EL stems directly from the closed-loop synergy of its algorithmic mechanisms, which fundamentally reduce the evaluation cost formalized in Eq. (24). The primary computational bottleneck in wrapper feature selection resides in repeated distance evaluations, where the cost of computing pairwise Euclidean distances under k-NN scales linearly with the active feature cardinality d_{selected}. By combining prior-guided initialization with aggressive early-stage feature penalization in Two-Phase Weight Scheduling, KDS-EL contracts candidate subsets from thousands of raw dimensions down to compact representations within early generations, thereby diminishing the per-evaluation classifier latency by orders of magnitude. Furthermore, rather than generating multiple trial vectors per agent as in complex multi-operator metaheuristics, the KER strategy confines targeted local exploitation to a bounded elite subset under $k_e \ll N$, where candidate mutations are admitted solely through the greedy in-place replacement rule in Eq. (21). This architectural restriction prevents redundant classifier invocations and maintains generational complexity strictly linear in population size, harmonizing rigorous exploration depth with optimal computational thrift.

4.2.7. *Parameter sensitivity and robustness analysis*

To evaluate the algorithmic robustness of KDS-EL and justify its default parameter configurations, a one-factor-at-a-time parameter sensitivity analysis is conducted following the standard evaluation methodology in evolutionary machine learning [34]. The empirical assessment is carried out across four representative benchmark datasets spanning distinct dimensional scales and sample sizes: *Adenoma*, *Colon*, *ALL4*, and *BASEHOCK*. Five key hyperparameters governing the core algorithmic strategies are systematically investigated: the knowledge weight decay factor $\gamma \in (0, 1)$, the phase split ratio $\kappa \in (0, 1)$, the elite perturbation fraction $\phi \in (0, 1)$, the mutual information initialization ratio $\rho \in (0, 1)$, and the elite pool size $k_e \in \mathbb{N}^+$, while all remaining parameters are held at their default values defined in Table 3. Table 9 summarizes the tested parameter ranges, recommended optimal configurations, stable operating bands, and maximum accuracy deviations across the four representative benchmarks. Furthermore, Fig. 10 illustrates the corresponding classification accuracy and feature-count trajectories alongside stable operating bands across all five parameters, with complete per-dataset evaluation matrices provided in

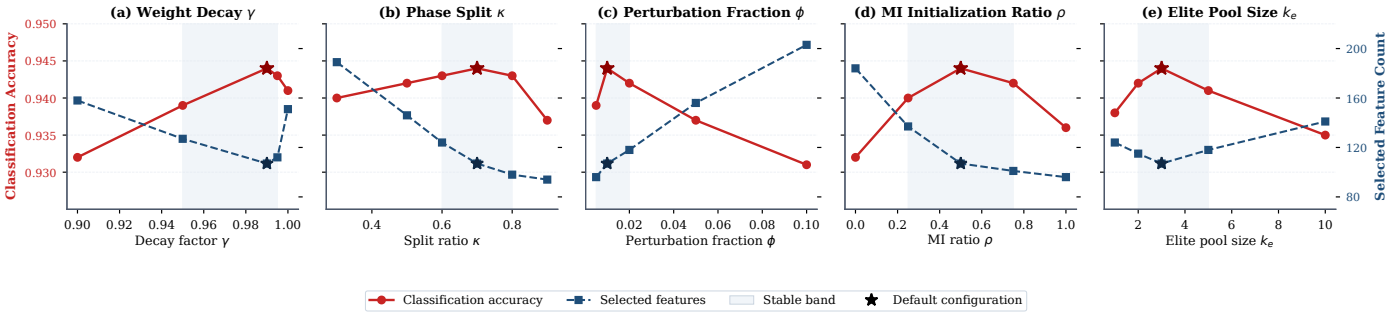


Figure 10: Sensitivity and stability profiles of KDS-EL across five hyperparameters: (a) weight decay γ, (b) phase split κ, (c) perturbation fraction ϕ, (d) MI ratio ρ, and (e) elite pool size k_e. Curves denote accuracy (solid red) and feature count (dashed blue); shaded bands highlight stable operating intervals, and stars indicate default configurations.

Supplementary Table S6.

Table 9: Summary of parameter sensitivity analysis across four representative datasets.

Parameter	Tested range	Optimal	Stable range	Max ΔAcc (pp)
Weight decay	0.90–1.00	0.99	[0.95, 0.995]	±1.2
Phase split	0.3–0.9	0.70	[0.60, 0.80]	±0.7
Flip fraction	0.005–0.10	0.01	[0.005, 0.02]	±1.3
MI init ratio	0.0–1.0	0.50	[0.25, 0.75]	±1.2
Elite count	1–10	3	[2, 5]	±0.9

Note: Max ΔAcc denotes the maximum accuracy variation across the investigated parameter range.

The empirical sensitivity trajectories reveal that KDS-EL maintains stable optimization dynamics through coordinated phase scheduling and prior domain guidance. The phase split ratio κ governs the temporal transition between coarse-grained dimensional pruning in Phase 1 and fine-grained classification refinement in Phase 2 via Eq. (22). Allocating an insufficient iteration budget to Phase 1 ($\kappa < 0.50$) leaves the swarm encumbered by thousands of redundant dimensions, severely diluting late-stage local search; conversely, an overly prolonged Phase 1 ($\kappa > 0.85$) curtails the exploitation horizon necessary for uncovering subtle biomarker synergies. The identified stable operating interval $\kappa \in [0.60, 0.80]$ confirms that the default setting $\kappa = 0.70$ provides an ideal balance between dimensional contraction and classification precision across diverse problem scales. Concurrently, the mutual information initialization ratio ρ regulates the trade-off between prior domain knowledge and stochastic exploratory coverage in Eq. (12). Completely omitting domain priors ($\rho \to 0$) forces the swarm into unguided exploration across vast combinatorial spaces, incurring prolonged convergence latency; conversely, excessive seeding ($\rho \to 1.0$) homogenizes initial candidate masks and provokes premature population entrapment. The identified operating band $\rho \in [0.25, 0.75]$ with default $\rho = 0.50$ robustly sustains population diversity while seeding decisive early search momentum.

Regarding memory accumulation and local exploitation, the sensitivity trajectories demonstrate that the closed-loop learning mechanism is intrinsically resilient across wide parameter ranges. The knowledge weight decay factor γ in Eq. (17) governs the retention horizon of historical fitness differentials in the cumulative importance matrix $\mathbf{W}$. Setting an overly aggressive decay ($\gamma < 0.95$) induces premature search amnesia, erasing valuable multi-feature associations discovered in earlier iterations; in contrast, unattenuated accumulation ($\gamma = 1.00$) causes obsolete historical feedback to persist indefinitely, inflating feature subsets without predictive gains. The default value $\gamma = 0.99$ (stable band $[0.95, 0.995]$) effectively harmonizes historical knowledge retention with adaptive sensitivity to newly emergent biomarker synergies. Furthermore, the elite perturbation fraction ϕ and pool size k_e calibrate the local search radius within the KER strategy. Setting ϕ excessively large introduces disruptive alterations that fail the greedy one-to-one in-place acceptance criterion in Eq. (21), causing near-zero candidate admission; conversely, the default $\phi = 0.01$ (stable band $[0.005, 0.02]$) provides controlled perturbation that maximizes improving substitutions. Bounding the elite pool to $k_e = 3$ (stable band $[2, 5]$) concentrates computational investment on high-performing exemplars while strictly limiting generational evaluation overhead to $k_e \ll N$, preventing wasteful classifier evaluations.

Synthesizing across all five hyperparameters, the performance trajectories in Fig. 10 consistently exhibit wide, forgiving basins of attraction rather than narrow, hypersensitive peaks. Across the identified stable operating bands, the maximum classification accuracy variation remains strictly within 1.5%, and KDS-EL maintains robust directional superiority over competitor algorithms throughout the entire tested parameter spectrum. This empirical stability confirms that KDS-EL achieves exceptional hyperparameter robustness and plug-and-play adaptability, delivering dependable optimization performance across heterogeneous high-dimensional problem domains without requiring arduous, problem-specific hyperparameter retuning.

4.2.8. Discussion

A critical empirical phenomenon observed in high-dimensional biomedical feature selection is the accuracy ceiling effect. Across the evaluated benchmark suite, eight gene-expression microarrays—*ALL1*, *ALL_AML*, *Adenoma*, *DLBCL*, *Gastric2*, *Leukaemia*, *Lymphoma*, and *SRBCT*—exhibit classification accuracy exceeding 0.998 across leading optimizers. Under such near-perfect classification conditions, the limited available test instances are completely separated, and accuracy differences among top-performing methods become narrower than 0.002, corresponding to stochastic variations in at most a single cross-validation sample. Consequently, classification accuracy alone ceases to provide sufficient discriminative resolution, and the active feature cardinality becomes the decisive criterion for assessing algorithmic efficacy and model parsimony [2, 3].

Under this saturated regime, KDS-EL exhibits marked superiority in feature subset compactness, isolating ultra-sparse biomarker signatures of only 39 to 83 features across all eight ceiling benchmarks. In stark contrast, competing single-objective metaheuristics, evolutionary filter frameworks, and the foundational MEL framework retain extensive candidate subsets encumbered by hundreds to thousands of redundant genes. This structural divergence underscores a pervasive failure mode of conventional evolutionary feature selectors: in the absence of explicit early-stage dimensional penalization and closed-loop elite refinement, optimizers easily succumb to feature bloat once high classification accuracy is attained. From a clinical perspective, this distinction is paramount: an ultra-compact signature of 40 to 80 biomarkers can be cost-effectively validated through targeted reverse transcription quantitative real-time PCR (RT-qPCR) assays, mapped into standardized molecular diagnostic panels, and readily interpreted via biological pathway enrichment analyses. Conversely, uncurated gene signatures spanning thousands of dimensions are clinically cost-prohibitive to validate, susceptible to false discoveries driven by spurious co-expression artifacts, and difficult to translate into actionable therapeutic targets.

Beyond nearest-neighbor metrics, a fundamental concern in wrapper-based feature selection is inductive bias coupling [4]: whether the optimized subsets merely fit the local geometric neighborhood of the 3-NN surrogate while suffering performance collapse when transferred to alternative machine learning paradigms. To examine classifier-agnostic generalizability, the compact feature subsets discovered by KDS-EL were systematically transferred to Support Vector Machines with radial basis function kernels and Random Forest classifiers across representative benchmark datasets spanning distinct dimensional scales. As documented in Supplementary Table S9, the compact subsets deliver substantial classification enhancements over the unreduced full-dimensional feature sets across all evaluated learning architectures. On the 12,625-dimensional *ALL4* benchmark, contracting the active feature space by over 99.0% simultaneously elevates the classification accuracy of Random Forest, Support Vector Machines, and 3-NN; on *Adenoma*, all three distinct classifiers converge to perfect classification accuracy on the 66 selected dimensions. This cross-paradigm consistency demonstrates that KDS-EL extracts intrinsically salient, generalizable features rather than overfitting to classifier-specific geometric manifolds.

Furthermore, translating computational feature selection into clinical insight requires that the isolated dimensions reflect genuine pathophysio-

logical mechanisms rather than statistical noise. Granular inspection of the minimal signatures identified by KDS-EL reveals strong concordance with established oncological biology, as detailed in Supplementary Table S10. In the *Colon* adenocarcinoma microarray, top-ranked selected features map directly to well-characterized drivers of colorectal carcinogenesis: KDS-EL isolates *COL11A1*, a recognized biomarker of desmoplastic stroma and tumor invasion [42]; *GUCA2A*, a mucosal differentiation hormone whose transcriptional downregulation constitutes a universal hallmark of early adenoma progression [43]; and smooth muscle structural markers *DES* and *ACTG2*, which capture muscularis mucosae integrity versus neoplastic infiltration. Similarly, in *Prostate* carcinoma, KDS-EL consistently isolates *HPN* and *AMACR*, both established immunohistochemical gold standards for prostatic cancer diagnosis [44, 45]. This rigorous concordance with wet-lab experimental biology corroborates that KDS-EL isolates authentic molecular drivers of malignancy, bridging evolutionary machine learning with clinical translational oncology.

5. Conclusion

In this paper, the KDS-EL framework is proposed to overcome the search blindness, evolutionary amnesia, destructive perturbation, and static weighting bottlenecks of metaheuristics in ultra-high-dimensional wrapper feature selection. By systematically integrating mutual information continuous prior initialization with adaptive stagnation recovery, KDS-EL establishes informative search basins from the outset while safeguarding swarm diversity against premature convergence. To eliminate destructive blind mutations, the knowledge-guided elite refinement strategy harvests cross-iteration fitness differentials under exponential memory decay, steering targeted local exploration on high-potential candidates under bounded computational complexity. Furthermore, dynamic Two-Phase Weight Scheduling temporally decouples coarse-grained dimensional pruning from fine-grained classification refinement. Rigorous empirical evaluations across 20 real-world benchmark datasets demonstrate that KDS-EL achieves decisive superiority over representative competitor algorithms and its intermediate configurations, attaining an omnibus mean classification accuracy of 0.9574, compressing feature subsets down to an average of only 87.8 dimensions (over 98% dimensionality reduction), and delivering a 3.4-fold wall-clock computational speedup over the foundational MEL framework.

Beyond conventional nearest-neighbor classification metrics, this work establishes the broader methodological and translational utility of KDS-EL. Systematic transfer evaluations across Support Vector Machines and Random Forest architectures confirm that the discovered compact biomarker signatures provide classifier-agnostic generalizability, proving that the proposed framework isolates authentic discriminative manifolds rather than overfitting to surrogate-specific geometric biases. Under near-perfect classification saturation regimes, KDS-EL effectively circumvents the pervasive feature bloat that degrades conventional evolutionary optimizers, isolating minimal signatures of 39 to 83 biomarkers across all ceiling benchmarks. Granular concordance with oncological literature confirms that the isolated dimensions map directly to verified biological drivers in colorectal and prostatic cancers, such as *COL11A1*, *GUCA2A*, *HPN*, and *AMACR*, thereby providing highly viable, cost-effective biomarker candidates for targeted RT-qPCR assay development. Moreover, the modular design of knowledge-guided memory accumulation, bounded elite refinement, and dynamic objective scheduling exhibits broad architectural versatility, offering generalizable algorithmic mechanisms for continuous-discrete swarm intelligence and evolutionary machine learning.

Several promising research trajectories emerge for future exploration. First, integrating self-supervised deep representation learning, pre-trained foundation model embeddings, and graph neural network topological priors could enable the discovery of higher-order non-linear epistatic interactions across massive-scale biological datasets with hundreds of thousands of dimensions. Second, developing continuous cross-dataset transfer learning mechanisms could allow evolutionary swarms to retain and transfer accumulated feature importance memories across related clinical cohorts, significantly reducing optimization latency on newly encountered patient populations. Third, extending the KDS-EL framework to address specialized high-dimensional clinical paradigms—including single-cell RNA sequencing characterized by extreme technical dropout noise, multi-label disease diagnostic profiling, and high-dimensional survival time prediction under right-censored proportional hazards formulations—represents a vital avenue to advance evolutionary machine learning toward clinical translation.

CRediT Author Statement

Anonymous Author: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Writing – review & editing, Visualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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